Digital Electronics

Detailed Solutions • 1992–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Digital Electronics (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7^{th} question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

Important Links & Resources

Visit our Website

prepfusion.in

Practice PYQs online

pyq.prepfusion.in

Subscribe: English Channel

youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel

youtube.com/@prepfusion-hindi

Join our Telegram group

t.me/All_About_Learning

Follow us on LinkedIn

linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal

youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha

youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A and Suresh Ravanam.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

SOLUTIONS

1

Logic Gates and Boolean Algebra

Topics

1. Introduction to Digital Systems
2. Basic Gates and Concept of Multivibrators Using NOT Gate
3. Universal and Arithmetic Gates
4. Implementing Boolean Expressions Using Universal Logic Gates
5. Timing Diagrams
6. Boolean Algebra, Venn Diagrams, and the Concept of Dual

Q1.25.1 MCQ 2025 1M Ans: B

The Boolean expression X is given as:

$$X = \bar{A}\bar{B} + \bar{A}\bar{C}$$

Factoring out $\bar{A}$ using the distributive law :

$$X = \bar{A}(\bar{B} + \bar{C})$$

Taking the complement of both sides:

$$\bar{X} = \overline{\bar{A}(\bar{B} + \bar{C})}$$

Applying De Morgan's theorem :

$$\bar{X} = \bar{\bar{A}} + \overline{(\bar{B} + \bar{C})}$$

Applying De Morgan's theorem again :

$$\bar{X} = A + BC \quad (\because \bar{\bar{x}} = x)$$

$$\boxed{(B)\ A + BC}$$

Key Points

- De Morgan's theorems: $\overline{x \cdot y} = \bar{x} + \bar{y}$ and $\overline{x + y} = \bar{x}\,\bar{y}$
- Distributive law : $xy + xz = x(y + z)$

Q1.23.1 MCQ 2023 1M Ans: D

For an XOR gate: $A \oplus B = \bar{A}B + A\bar{B}$. Evaluating each expression:

(i) $X \oplus X$:

$$X \oplus X = \bar{X}X + X\bar{X} = 0 + 0 = 0 \quad \rightarrow \mathbf{q}$$

(ii) $X \oplus \bar{X}$:

$$X \oplus \bar{X} = \bar{X} \cdot \bar{X} + X \cdot X = \bar{X} + X = 1 \quad \rightarrow \mathbf{p}$$

(iii) $X \oplus 0$:

$$X \oplus 0 = \bar{X} \cdot 0 + X \cdot \bar{0} = 0 + X \cdot 1 = X \quad \rightarrow \mathbf{s}$$

(iv) $X \oplus 1$:

$$X \oplus 1 = \bar{X} \cdot 1 + X \cdot \bar{1} = \bar{X} + 0 = \bar{X} \quad \rightarrow \mathbf{r}$$

The correct matching is: (i)–q, (ii)–p, (iii)–s, (iv)–r.

$$\boxed{(D)\ \text{(i) - q , (ii) - p , (iii) - s , (iv) - r}}$$

Key Points

- $X \oplus 0 = X$ makes XOR useful as a *controlled buffer*: the output equals the input when the control is 0.
- $X \oplus 1 = \bar{X}$ makes XOR useful as a *controlled inverter*: the output is the complement of the input when the control is 1.

Mistakes to be avoided

- Verify using the definition $A \oplus B = \bar{A}B + A\bar{B}$; substituting directly avoids sign or logic errors.

Q1.23.2 MCQ 2023 2M Ans: A

Given circuit is shown below with propagation delay of gates mentioned.

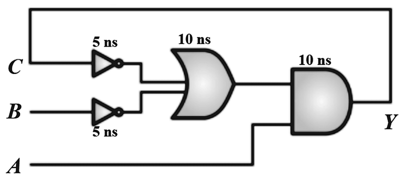
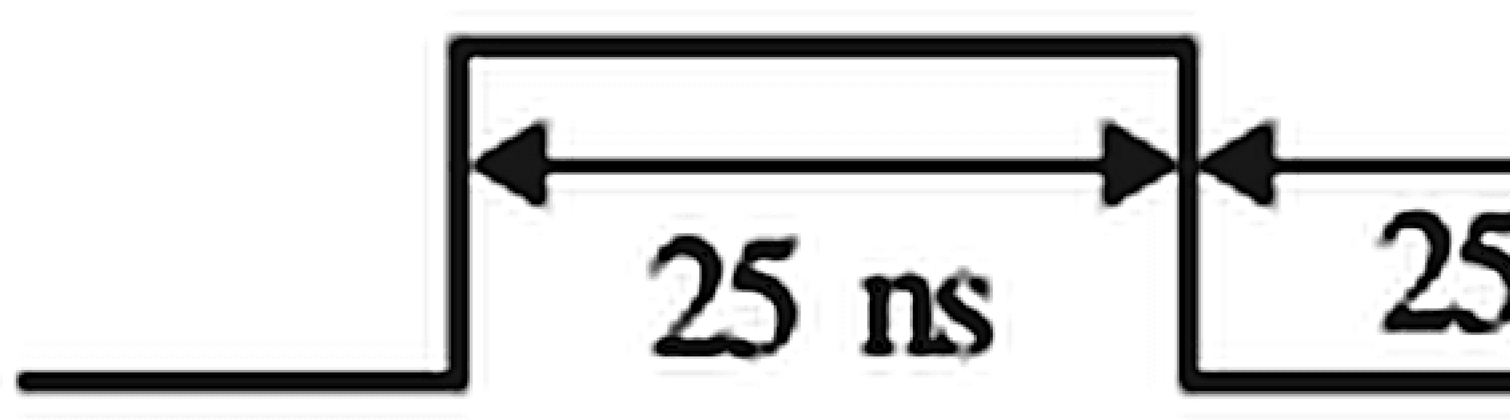


Fig. Solution Q1.23.2

Since $A = B = 1$, and C is 1 initially, $Y = 0$. This value of Y is sent as feedback to C, and then C goes to 0, causing Y to go to 1. Thus Y oscillates between 0 and 1.

Total propagation delay from C to Y = 5 ns + 10 ns + 10 ns = 25 ns.

Thus Y toggles between 0 and 1 every 25 ns, as shown:



Time period of output $T = 50ns$. Thus frequency of Y will be

$$f = \frac{1}{T} = \frac{1}{50\text{ns}} = 20\text{MHz}$$

$$\boxed{(A)\ 20\text{MHz}}$$

Q1.22.1 MCQ 2022 2M Ans: D

The output F of the given logic circuit is:

$$F = \overline{\overline{\bar{A} + \bar{B}} + \overline{A + \bar{B}}}$$

By De-Morgan's Law,

$$F = \overline{AB + \bar{A}B} = \overline{B}$$

$$\boxed{(D)\ \overline{B}}$$

Q1.21.1

MCQ | 2021 | 1M | Ans: D

Method 1

Given:

$$F(x,y,z) = (x' + y + z)(x + y' + z')(x' + y + z')(x'y'z' + x'yz' + xyz')$$

Rearranging the terms:

$$F(x,y,z) = (x' + y + z)(x' + y + z')(x + y' + z')(x'y'z' + x'yz' + xyz')$$

$$F = (x' + y + z)(x' + y + z')(x + y' + z')\{(x'y'z' + x'yz') + (x'yz' + xyz')\} \quad [\because x = x + x]$$

Taking $x'z'$ common in the last term, and $x' + y$ from the first two terms:

$$F = (x' + y)(z + z')(x + y' + z')\{x'z'(y' + y) + yz'(x' + x)\}$$

$$F = (x' + y)(x + y' + z')(x'z' + yz')$$

Taking z' common in the last term:

$$F = (x' + y)(x + y' + z')(x' + y)z' = (x + y' + z')(x' + y)z'$$

Applying absorption law $A(A + B) = A$:

$$F = (x' + y)z' = x'z' + yz'$$

Method 2

Given:

$$F(x,y,z) = (x' + y + z)(x + y' + z')(x' + y + z')(x'y'z' + x'yz' + xyz')$$

After multiplication and simplification,

$$F(x,y,z) = x'y'z' + x'yz' + xyz' = x'z' + xyz'$$

$$F(x,y,z) = z'(x' + xy) = z'(x' + y) = x'z' + yz'$$

$$\boxed{(D)\ F(x,y,z) = x'z' + yz'}$$

Key Points

- Absorption law: $A(A + B) = A$

Q1.20.1

MCQ | 2020 | 1M | Ans: D

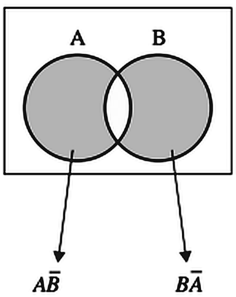


Fig. Solution Q1.20.1

Thus the Boolean expression is:

$$F(A,B) = \bar{A}B + A\bar{B}$$

Converting to POS form.

$$F(A,B) = (\bar{A} + A\bar{B})(B + A\bar{B}) = (\bar{A} + \bar{B})(A + B)$$

This corresponds to option (D).

$$\boxed{(D)\ (A+B).(\bar{A}+\bar{B})}$$

Q1.19.1 NAT 2019 1M Ans: 256 ● ○ ○

Total number of Boolean functions over n Boolean variables is $= 2^{2^n}$. Here $n = 3$, therefore, number of Boolean functions $= 2^{2^3} = 2^8 = 256$.

$$\boxed{256}$$

Q1.18.1 MCQ 2018 2M Ans: A ● ○ ○

The given circuit is XOR gate realised using NAND gates.
Proof:

$$F = \overline{\overline{X.\overline{XY}}.\overline{Y.\overline{XY}}} = X.\overline{XY} + Y.\overline{XY}$$

$$F = \bar{X}Y + X\bar{Y}$$

$$\boxed{(A)\ \bar{X}Y + X\bar{Y}}$$

Q1.17.1 MCQ 2017 2M Ans: C ● ○ ○

The logic realised by the circuit is:

$$X = AB + \bar{B}\bar{A}$$

Hence, C is the correct option.

$$\boxed{(C)\ AB + \bar{B}\bar{A}}$$

Q1.16.1 MCQ 2016 1M Ans: C ● ○ ○

The given function can be simplified as:

$$f = XY + (X' + Y')Z = XY + X'Z + Y'Z$$

$$f = (XY + X')(XY + Z) + (XY + Y')(XY + Z) \quad [\because\ A + BC = (A+B)(A+C)]$$

$$f = (X' + Y)(XY + Z) + (X + Y')(XY + Z) \quad [\because\ A' + AB = A' + B]$$

$$f = (XY + Z)(X + Y' + X' + Y) \quad [\because\ AC + BC = (A+B)C]$$

$$f = XY + Z = (X+Z)(Y+Z)$$

$$\boxed{(C)\ (X+Z)(Y+Z)}$$

Q1.15.1 | MCQ | 2015 | 2M | Ans: D

In the given circuit, the intermediate nodes are marked as A, B, C .

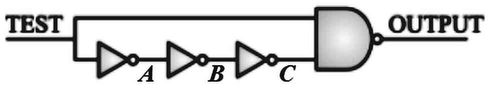
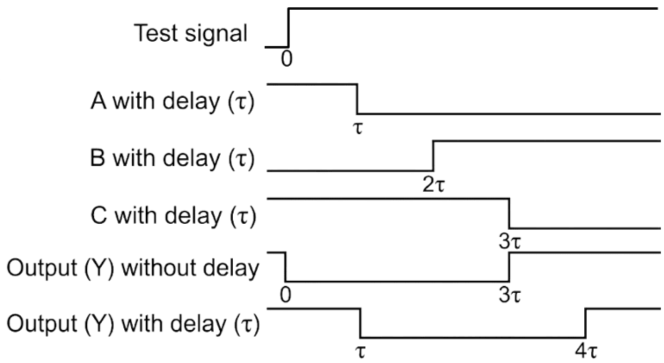


Fig. Solution Q1.15.1

It is known that

- TEST was initially at logic LOW and then switched to logic HIGH.
- All gates have the same non-zero propagation delay (Let's asssume τ).

For given transition of TEST, the transition diagram for the circuit can be drawn as:



Thus, it is observed that OUTPUT signal is initially HIGH, goes LOW, and returns to HIGH after a while.

$$\boxed{(D)\text{ Pulses from HIGH to LOW to HIGH}}$$

Q1.15.2 | MCQ | 2015 | 2M | Ans: A

The logic obtained from the circuit is $X\bar{Y} + Y\bar{X}$. Hence, A is the correct option.

$$\boxed{(A)\ X\bar{Y} + Y\bar{X}}$$

Q1.13.1 | MCQ | 2013 | 2M | Ans: C

First Case:

Let the initial condition be that both switches S_1 and S_2 are ON and bulb Y is also ON.

$$\text{i.e. ,}\quad \begin{bmatrix} S_1 = 1 \implies ON \\ S_2 = 1 \implies ON \end{bmatrix} \implies Y = 1 \implies \text{Bulb } ON$$

Based on this initial condition, the functional table can be made as:

From this table, it is clear that $Y = S_1 \odot S_2$. Hence the correct option is C .

Second Case:

Let the initial condition be that both switches S_1 and S_2 are OFF and bulb Y is also OFF.

$$\text{i.e. ,}\quad \begin{bmatrix} S_1 = 1 \implies OFF \\ S_2 = 1 \implies OFF \end{bmatrix} \implies Y = 0 \implies \text{Bulb } OFF$$

Based on this initial condition, the functional table can be made as:

From this table, it is clear that $Y = S_1 \oplus S_2$ which does not match any option .

$$\boxed{(C)\text{ XNOR}}$$

Switches		Bulb	States
S_1	S_2	Y	
1 (ON)	1 (ON)	1 (ON)	Initially bulb is ON
↓ As it is	Press ↓	↓	
1 (ON)	0 (OFF)	0 (OFF)	Bulb is OFF because previously it was ON
Press ↓	↓ As it is	↓	
0 (OFF)	0 (OFF)	1 (ON)	Bulb is ON because previously it was OFF
↓ As it is	Press ↓	↓	
0 (OFF)	1 (ON)	0 (OFF)	Bulb is OFF because previously it was ON

Switches		Bulb	States
S_1	S_2	Y	
0 (OFF)	0 (OFF)	0 (OFF)	Initially bulb is OFF
↓ As it is	Press ↓	↓	
0 (OFF)	1 (ON)	1 (ON)	Bulb is ON because previously it was OFF
Press ↓	↓ As it is	↓	
1 (ON)	1 (ON)	0 (OFF)	Bulb is OFF because previously it was ON
↓ As it is	Press ↓	↓	
1 (ON)	0 (OFF)	1 (ON)	Bulb is ON because previously it was OFF

Key Points

- This kind of circuit can be realised using both XOR and XNOR gates, but in options only XNOR is mentioned, so that is the correct option.

Q1.10.1 MCQ 2010 2M Ans: A

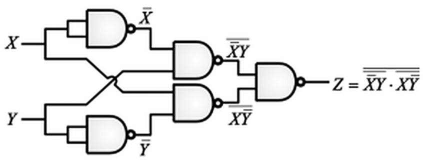


Fig. Solution Q1.10.1

Solving, $Z = \overline{\overline{\overline{X}Y}.\overline{\overline{Y}X}} = \overline{X}Y + X\overline{Y} = X \oplus Y$

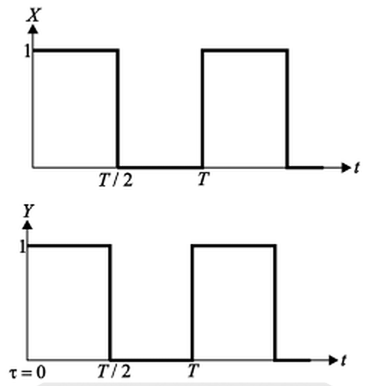
$\boxed{(A)\ \text{XOR}}$

Q1.07.1 MCQ 2007 2M Ans: C ● ● ○

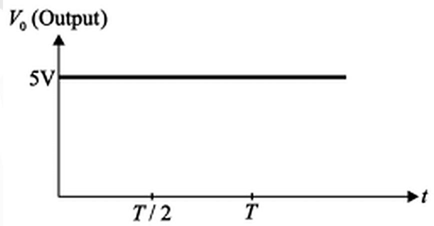
From the given truth table it is clear that the digital circuit mentioned is an XNOR gate. Let's consider 3 cases to check dependency of average value of output on τ .

Case I: $\tau = 0$

The inputs are:



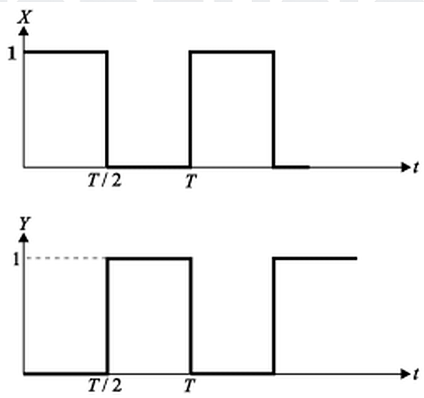
Corresponding output is:



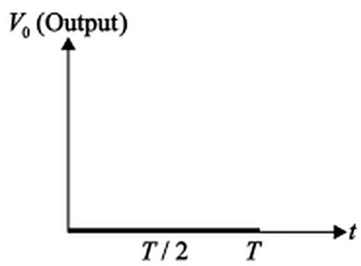
Here, average value of output is $V_{avg} = 5V$.

Case II: $\tau = \dfrac{T}{2}$

The inputs are:



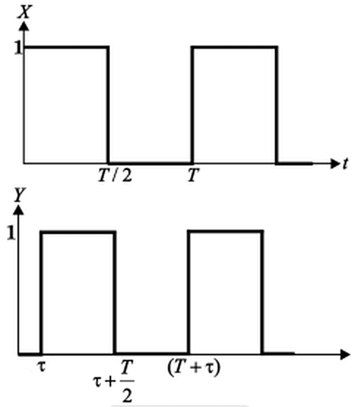
Corresponding output is:



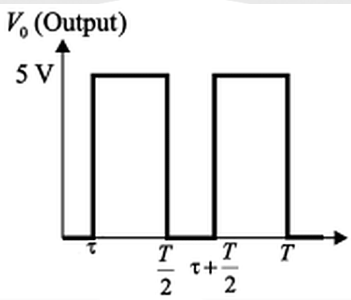
Here, average value of output is $V_{avg} = 0V$.

Case III: $O < \tau < \frac{T}{2}$

The inputs are:



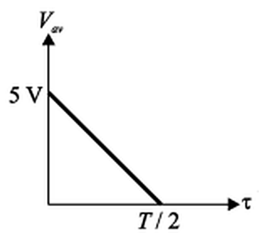
Corresponding output is:



Here, average value of output is

$$V_{avg} = \frac{5\left(\frac{T}{2} - \tau\right)}{\frac{T}{2}} = 5\left(1 - \frac{2\tau}{T}\right)$$

Thus dependency of average value of output on τ can be shown as:



$\boxed{(C)}$

Q1.05.1 MCQ 2005 1M Ans: A

Given:

$$Y = \overline{AB}$$

Applying De-Morgan's Law,

$$Y = \overline{AB} = \bar{A} + \bar{B}$$

which matches option A.

$$\boxed{(A)\ \bar{A} + \bar{B}}$$

Q1.05.2 MCQ 2005 1M Ans: D

- Subtraction can be performed using 2's complement arithmetic: $A - B = A + \bar{B} + 1$
- A bank of K XOR gates is used to conditionally complement each bit of B.
- A control signal M is applied to all XOR gates and also used as the initial carry input. When M = 0, the circuit performs addition. When M = 1, the circuit performs subtraction.

This is why a K-bit full adder together with K XOR gates is sufficient to realize both operations.

$$\boxed{(D)\ \text{K 2-Input XOR Gates}}$$

Q1.02.1 MCQ 2002 1M Ans: B

A ring oscillator uses an odd number of inverters connected in a feedback loop. The fundamental output frequency is given by:

$$f_0 = \frac{1}{2Nt_{pd}}$$

where N = number of inverters = 5 and t_{pd} = propagation delay of each inverter.
Given $f_0 = 10\,\text{MHz} = 10 \times 10^6$ Hz, solving for t_{pd}:

$$t_{pd} = \frac{1}{2Nf_0} = \frac{1}{2 \times 5 \times 10 \times 10^6}$$

$$t_{pd} = \frac{1}{10^8} = 10^{-8}\,\text{s} = 10\ \text{ns}$$

$$\boxed{(B)\ 10\ \text{ns}}$$

Key Points

- The oscillation period is $T = 2Nt_{pd}$; each inverter contributes one propagation delay t_{pd} per half-cycle, and the signal traverses all N inverters twice per full cycle.

Q1.01.1 MCQ 2001 1M Ans: C

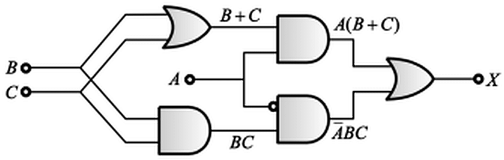


Fig. Solution Q1.01.1

Thus final output of the given circuit is:

$$X = A(B+C) + \bar{A}BC = AB + AC + \bar{A}BC = AB + C(A + \bar{A}B)$$

$$X = AB + C(A+B) = AB + BC + CA$$

$$\boxed{(C)\ AB + BC + CA}$$

Q1.00.1 MCQ 2000 1M Ans: A

Let Y be a Boolean function such that

$$Y = A + \bar{A}B = (A+\bar{A})(A+B) = A + B$$

$$\boxed{(A)\ A + B}$$

Q1.98.1 MCQ 1998 1M Ans: A

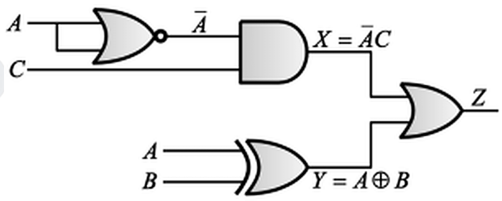


Fig. Solution Q1.98.1

From the given circuit

$$X = \bar{A}.C$$

$$Y = A \oplus B$$

Thus final output Z is

$$Z = X + Y = \bar{A}C + \bar{A}B + A\bar{B}$$

$$\boxed{(A)\ \bar{A}C + A\bar{B} + \bar{A}B}$$

Q1.95.1 MCQ 1995 1M Ans: B ● ○ ○

If A and B are inputs to a logic gate, then the logic according to its output is shown:

NOR	$\bar{A}.\bar{B}$
NAND	$\bar{A}+\bar{B}$
XNOR	$\bar{A}\bar{B}+AB$
AND	$\overline{\bar{A}+\bar{B}}$

The final result is :

$$\boxed{(B)\ \text{a - ii , b - iv , c - iii , d - i}}$$

Q1.94.1 MCQ 1994 1M Ans: B ● ○ ○

Applying De-Morgan's law,

$$Y = \overline{A+B+C} = \bar{A}.\bar{B}.\bar{C}$$

$$\boxed{(B)\ \bar{A}.\bar{B}.\bar{C}}$$

Q1.94.2 MCQ 1994 1M Ans: A ● ○ ○

NAND gate can be used to realise any logic function since it is a universal gate.

$$\boxed{(A)\ \text{NAND gate}}$$

Q1.92.1 MCQ 1992 1M Ans: MTA ● ○ ○

Given circuit, for $A = 0$:

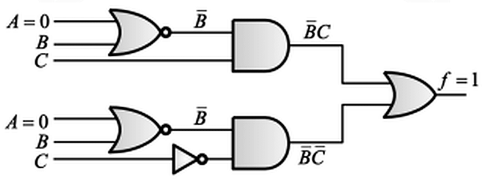


Fig. Solution Q1.92.1

Output function $f = \bar{B}C + \bar{B}\bar{C} = \bar{B}$.
Therefore for output to be 1, $B = 0$ and C may be 0 or 1 since output does not depend on C .
However, this does not match any of the given options, so it is marked as an MTA question.

$$\boxed{\text{MTA}}$$

DEBUG INFO

Chapter I

Logic Gates and Boolean Algebra

Total Questions : 25

MCQ : 24

MSQ : 0

NAT : 1

SOLUTIONS

II

Representation of Boolean Expressions and K-Maps

Topics

1. Minterms and Maxterms
2. SOP and POS Representation
3. Karnaugh Maps (K-Maps)
4. Concept of Implicants

Q2.24.1 MCQ 2024 1M Ans: A

Method 1

The given expression is:

$$F = (A + \bar{A}B) + \bar{A}(A + \bar{B})C$$

Using distributive and redundancy laws:

$$F = (A + \bar{A})(A + B) + (\bar{A}A + \bar{A}\bar{B})C = (A + B) + (0 + \bar{A}\bar{B})C = A + B + \bar{A}\bar{B}C$$

Expanding $\bar{A}\bar{B}C$ and regrouping:

$$F = A + \bar{A}\bar{B}C + B + \bar{A}\bar{B}C = A + \bar{B}C + B + \bar{A}C = A + C + B + C$$

$$F = A + B + C$$

Method 2

Expanding the given expression to collect minterms:

$$F = (A + \bar{A}B) + \bar{A}(A + \bar{B})C = A + \bar{A}B + \bar{A}\bar{B}C$$

$$F = A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC + \bar{A}B\bar{C} + \bar{A}BC + \bar{A}\bar{B}C$$

$$F = \Sigma m(1, 2, 3, 4, 5, 6, 7)$$

The K-Map is given as:

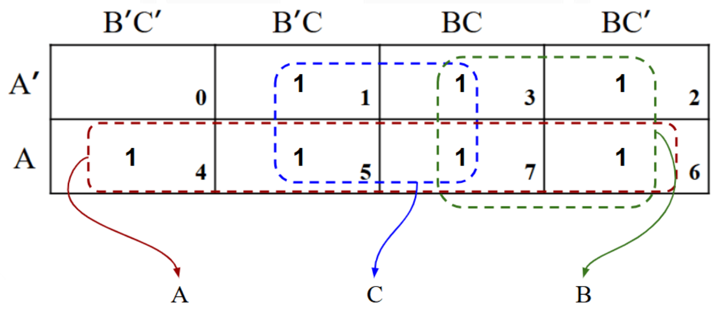


Fig. Solution Q2.24.1

From the K-Map it is observed that the minterms corresponding to the groups are A, B and C. Thus we get the Boolean expression as

$$F = A + B + C$$

$$\boxed{(A)\ A + B + C}$$

Key Points

- Redundancy law $A + \bar{A}C = A + C$ (also called Absorption Law Variant).
- $A + BC = (A + B)(A + C)$ is a variant of Distribution Law.
- OR complement law : $A + \bar{A} = 1$
- AND complement law : $A.\bar{A} = 0$

Q2.23.1 MCQ 2023 1M Ans: C

The minterms of the Boolean expression are given as:

$$F(W, X, Y, Z) = \Sigma(4, 5, 10, 11, 12, 13, 14, 15)$$

To determine the minimized expression, we draw the K-Map:

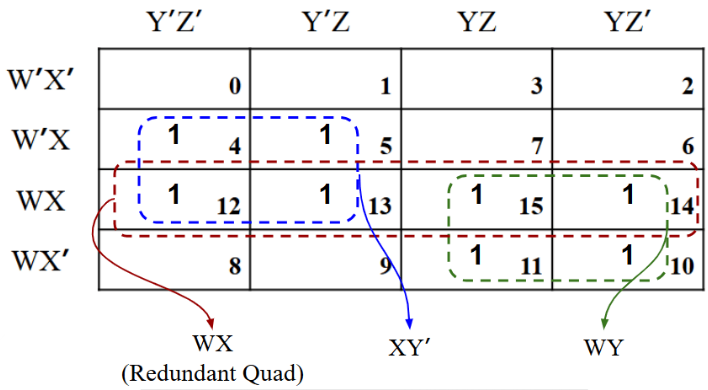


Fig. Solution Q2.23.1

Thus the minimzed SOP expression obtained is:

$$F(W, X, Y, Z) = X\bar{Y} + WY$$

$$\boxed{(C)\ X\bar{Y} + WY}$$

Mistakes to be avoided

- The WX group obtained in the K-Map is redundant, i.e., every minterm in that group is already part of a different group. Redundant groups must be ignored while writing the final expression.

Q2.22.1 MCQ 2022 1M Ans: B

For 2-bit unsigned binary numbers $A = a_1a_0$ and $B = b_1b_0$, the function $F(A > B)$ is defined as:

$$F(a_1, a_0, b_1, b_0) = \begin{cases} 1, & A > B \\ 0, & \text{otherwise} \end{cases}$$

The truth table is:

a_1	a_0	b_1	b_0	$F(A > B)$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

From the truth table, $F(A > B) = 1$ for the following minterms:

$$F(A > B) = \sum m(4, 8, 9, 12, 13, 14)$$

Plotting on a 4-variable K-map with variables $a_1 a_0$ (rows) and $b_1 b_0$ (columns):

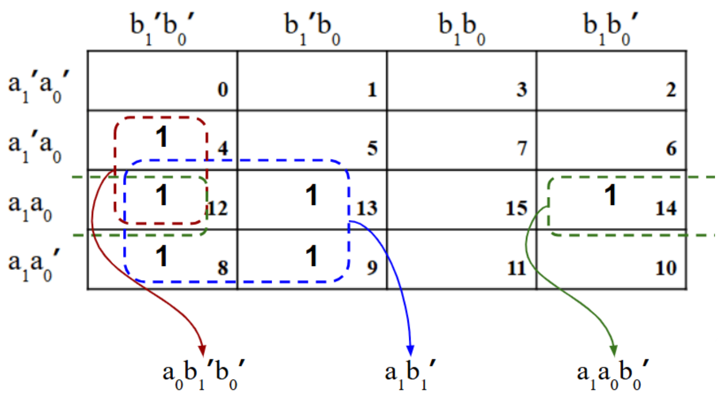


Fig. Solution Q2.22.1

Grouping the 1s, the minimised SOP expression obtained is:

$$F(A > B) = a_1\bar{b}_1 + a_0\bar{b}_1\bar{b}_0 + a_1 a_0\bar{b}_0$$

$$\boxed{(B)\ a_1\bar{b}_1 + a_0\bar{b}_1\bar{b}_0 + a_1 a_0\bar{b}_0}$$

Key Points

- For an n-bit comparator: total combinations $= 2^{2n}$; combinations where $A = B$ is 2^n; combinations where $A > B$ (or $A < B$) $= (2^{2n} - 2^n)/2$.
- The SOP expression for $A > B$ in a 2-bit comparator reflects the MSB comparison first ($a_1\bar{b}_1$), then LSB comparison when MSBs are equal ($a_1 a_0\bar{b}_0$ and $a_0\bar{b}_1\bar{b}_0$).

Mistakes to be avoided

- Do not miss the wrap-around adjacency in the K-map; minterms in the top and bottom rows, or leftmost and rightmost columns, are adjacent.

Q2.20.1 | MCQ | 2020 | 1M | Ans: C

The given Function is:

$$f(A,B,C,D) = \Sigma m(0,1,2,6,8,9,10,11) + \Sigma d(3,7,14,15)$$

The K-Map can be drawn as:

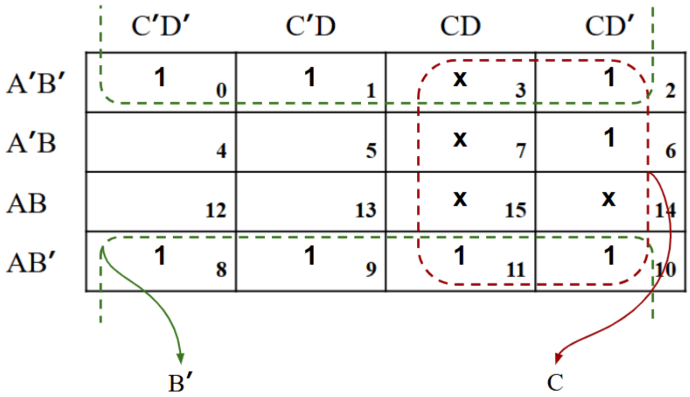


Fig. Solution Q2.20.1

Thus the minimal SOP expression obtained will be:

$$f = \bar{B} + C$$

$$\boxed{(C)\ \bar{B} + C}$$

Key Points

- Don't-care terms should be clubbed together with minterms to make larger groups.

Mistakes to be avoided

- Do not create a group of only don't-care terms since that will be completely redundant.

Q2.18.1 | MCQ | 2018 | 1M | Ans: B

The function F is given as:

$$F(A,B,C) = (A+B+\bar{C}).(A+\bar{B}+\bar{C}).(\bar{A}+B+C).(\bar{A}+\bar{B}+\bar{C})$$

Since it is a POS expression, it can be written with maxterms as:

$$F(A,B,C) = \pi M(001,011,100,111) = \pi M(1,3,4,7)$$

Converting to SOP form,

$$F(A,B,C) = \Sigma m(0,2,5,6) = \Sigma m(000,010,101,110)$$

Thus the SOP expression is obtained as:

$$F(A,B,C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$$

$$\boxed{(B)\ \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}}$$

Key Points

- To obtain minterms from maxterms, identify the indices of the missing terms relative to the total possible combinations for the given number of variables ; those are the minterms

Q2.12.1 MCQ 2012 1M Ans: A

The function is given as:

$$f(X,Y,Z) = \Sigma m(2,3,4,5)$$

In SOP form:

$$f(X,Y,Z) = \bar{X}Y\bar{Z} + \bar{X}YZ + X\bar{Y}\bar{Z} + X\bar{Y}Z$$

The K-Map is:

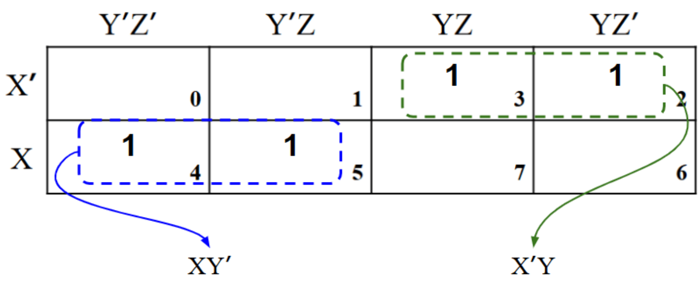


Fig. Solution Q2.12.1

Thus the prime implicants obtained are:

$$XY' \text{ and } X'Y$$

$$\boxed{(A)\ XY'\ ,\ X'Y}$$

Q2.11.1 MCQ 2011 1M Ans: A

The function is given as:

$$f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$$

In minterm form,

$$f = \Sigma m(0,2,4,6,7)$$

Thus maxterms are:

$$f = \pi M(1,3,5)$$

The POS K-Map is obtained as:

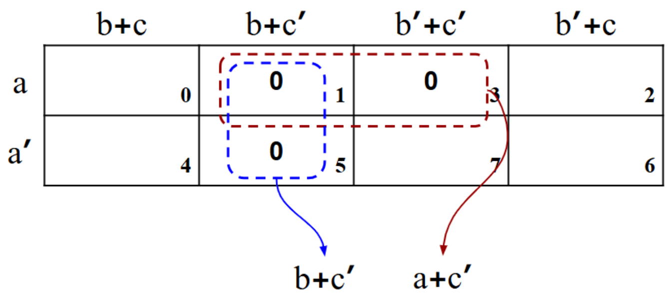


Fig. Solution Q2.11.1

Thus the final POS expression is:

$$f = (b + \bar{c})(a + \bar{c})$$

$$\boxed{(A)\ (b + \bar{c})(a + \bar{c})}$$

Q2.09.1 | MCQ | 2009 | 1M | Ans: D

In the given K-Map, groups can be formed as shown:

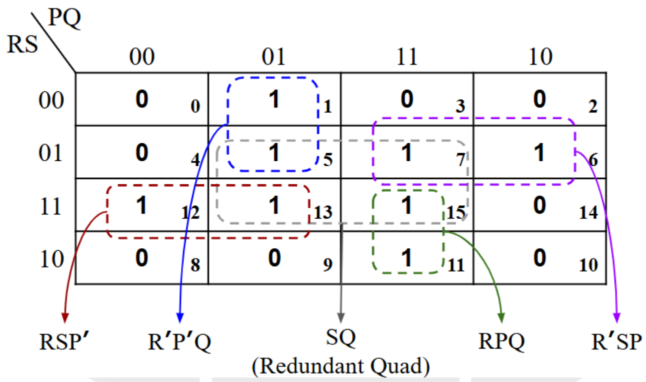


Fig. Solution Q2.09.1

Ignoring the redundant quad, the SOP expression can be written as:

$$f = P\bar{R}S + PQR + \bar{P}RS + \bar{P}Q\bar{R}$$

$$\boxed{(D)\ P\bar{R}S + PQR + \bar{P}RS + \bar{P}Q\bar{R}}$$

Q2.08.1 | MCQ | 2008 | 1M | Ans: A

Method 1

Given $y = \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}S + P\bar{Q}RS + P\bar{Q}R\bar{S} + \bar{P}\bar{Q}R\bar{S}$

$$y = \bar{Q}\bar{R}\bar{S}(P + \bar{P}) + P\bar{Q}S(R + \bar{R}) + \bar{Q}R\bar{S}(P + \bar{P}) = \bar{Q}\bar{R}\bar{S} + P\bar{Q}S + \bar{Q}R\bar{S}$$

$$y = \bar{Q}\bar{S}(\bar{R} + R) + P\bar{Q}S = \bar{Q}\bar{S} + P\bar{Q}S = \bar{Q}(\bar{S} + PS) = \bar{Q}(\bar{S} + P)$$

$$y = \bar{Q}\bar{S} + P\bar{Q}$$

Method 2

Given $y = \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}S + P\bar{Q}RS + P\bar{Q}R\bar{S} + \bar{P}\bar{Q}R\bar{S}$.
The expression can be written in minterm form as follows:

$$y = \Sigma m(0, 2, 8, 9, 10, 11)$$

The K-Map is:

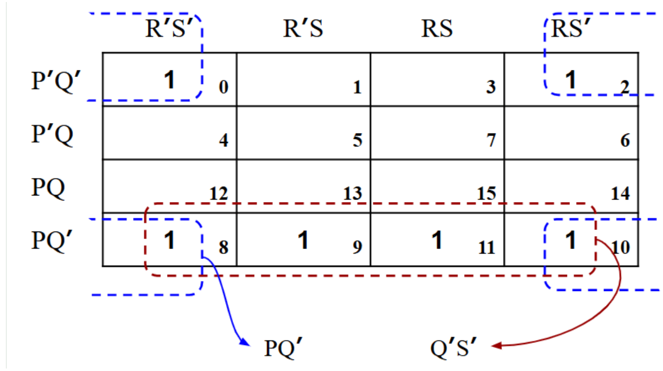


Fig. Solution Q2.08.1

Thus, minimized SOP expression is obtained as $y = \bar{Q}\bar{S} + P\bar{Q}$

$$\boxed{(A)\ \bar{Q}\bar{S} + P\bar{Q}}$$

Mistakes to be avoided

- Make sure to check the 4 corners of the K-Map, since those terms may also form a quad.

Q2.07.1

MCQ | **2007** | **1M** | **Ans: D** | ● ○ ○

The function is

$$f = \bar{x}.y + x.\bar{y}.\bar{z}$$

Expanding to obtain canonical SOP form:

$$f = \bar{x}.y.(z + \bar{z}) + x.\bar{y}.\bar{z} = \bar{x}.y.z + \bar{x}.y.\bar{z} + x.\bar{y}.\bar{z}$$

The minterm indices are:

$$f = \Sigma m(2, 3, 4)$$

It is also mentioned that the input combination $x = y = 1$ can never occur. Thus minterms 110 and 111 are invalid and must be considered as don't-cares. Thus the final minterms are:

$$f = \Sigma m(2, 3, 4) + \Sigma d(6, 7)$$

The K-Map is drawn as:

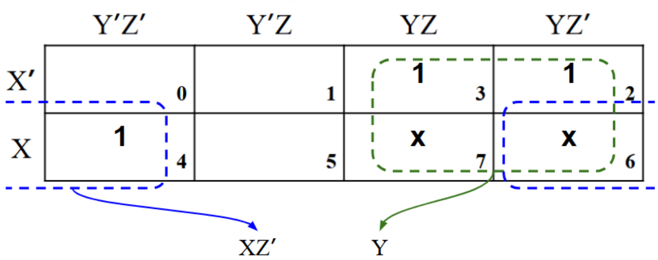


Fig. Solution Q2.07.1

The final minimal SOP expression is $f = x\bar{z} + y$

$$\boxed{(D)\ y + x.\bar{z}}$$

Key Points

- Canonical SOP (also called Sum of Minterms) is the complete, unsimplified representation where every term contains all input variables of the function, either in complemented or uncomplemented form.

Q2.06.1 MCQ 2006 1M Ans: C

Given:

$$f(A, B, C) = \Sigma m(0, 1, 2, 3, 5, 6)$$

K-Map is drawn as:

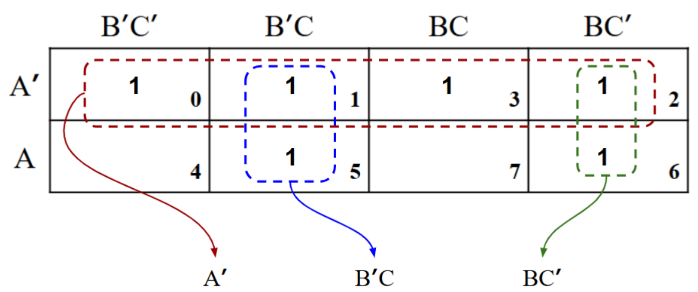


Fig. Solution Q2.06.1

The SOP obtained from K-Map can be simplified as follows:

$$f(A, B, C) = \bar{A} + \bar{B}C + B\bar{C} = \bar{A} + (B \oplus C)$$

$$\boxed{(C)\ \bar{A} + (B \oplus C)}$$

Q2.05.1 MCQ 2005 1M Ans: C

With the data given, the truth table can be made as:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

The minterms of Y are obtained as:

$$Y = \Sigma m(3, 5, 6, 7)$$

The K-Map is drawn as:

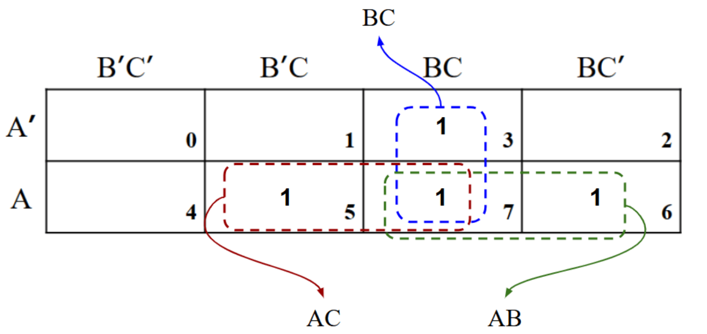


Fig. Solution Q2.05.1

The minimized SOP expression is

$$Y = AB + BC + AC$$

Thus the maximum number of 2-input terms is 3.

$$\boxed{(C)\ 3}$$

Q2.04.1 MCQ 2004 1M Ans: D

Method 1

Let the Boolean function be written as $F = ABC\bar{D} + ABC\bar{D}$...

$$F = AB\bar{C}(\bar{D} + D) + ABC(\bar{D} + D) = AB\bar{C} + ABC = AB(\bar{C} + C) = AB$$

Method 2

Writing the minterm indices of the given Boolean function:

$$F = \Sigma m(12, 13, 14, 15)$$

The K-Map is:

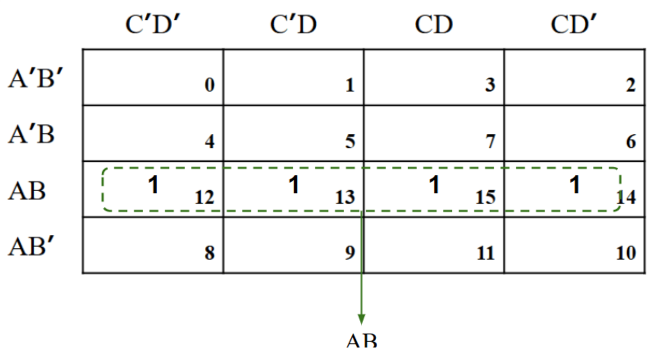


Fig. Solution Q2.04.1

Thus the final expression is $F = AB$.

$$\boxed{(D)\ AB}$$

Q2.03.1 | MCQ | 2003 | 1M | Ans: B

Creating groups in the K-Map given:

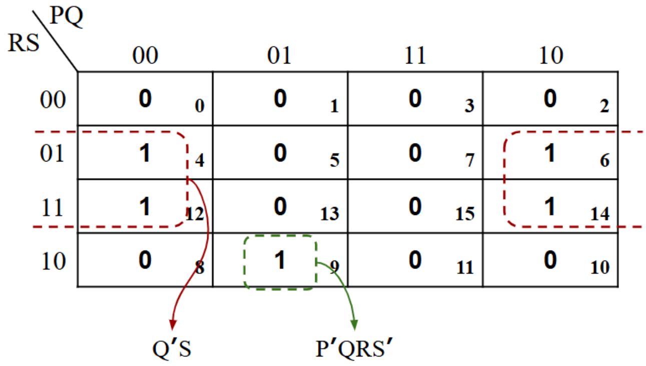


Fig. Solution Q2.03.1

The SOP expression obtained is $\bar{P}QR\bar{S} + \bar{Q}S$.

$$\boxed{(B)\ \bar{P}QR\bar{S} + \bar{Q}S}$$

Q2.00.1 | MCQ | 2000 | 1M | Ans: A

From the given table, the canonical SOP form is:

$$Q = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

Method 1

Simplifying, we get:

$$Q = AB(\bar{C} + C) + BC(\bar{A} + A) + CA(\bar{B} + B) = AB + BC + CA$$

Method 2

Simplifying, we get:

$$Q = AB(\bar{C} + C) + \bar{A}BC + A\bar{B}C = AB + \bar{A}BC + A\bar{B}C$$

$$Q = A(\bar{B}C + B) + \bar{A}BC = AB + AC + \bar{A}BC$$

$$Q = B(A + \bar{A}C) + AC = AB + BC + CA$$

Method 3

K-Map can be drawn as:

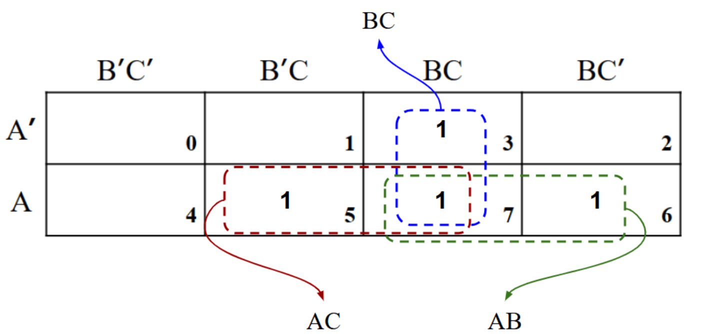


Fig. Solution Q2.00.1

The obtained SOP expression is

$$Q = AB + BC + AC$$

$$\boxed{(A)\ AB + BC + CA}$$

Q2.98.1 MCQ 1998 1M Ans: B ● ○ ○

The function given is $f = A\bar{B}CD + \bar{A}BC + BCD + \bar{A}\bar{B}C$.

Method 1

The function can be minimized as:

$$f = CD(A\bar{B} + \bar{A}) + \bar{A}C(B + \bar{B}) + BCD = CD(\bar{A} + \bar{B}) + \bar{A}C + BCD$$

$$f = \bar{A}CD + \bar{A}C + \bar{B}CD + BCD = \bar{A}C + CD$$

Method 2

The function can be written in canonical SOP form as

$$f = A\bar{B}CD + \bar{A}BC\bar{D} + \bar{A}BCD + \bar{A}BCD + ABCD + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D}$$

Removing repeated terms,

$$f = A\bar{B}CD + \bar{A}BC\bar{D} + \bar{A}BCD + ABCD + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D}$$

The minterm indices are:

$$f = \Sigma m(2, 3, 6, 7, 11, 15)$$

The K-Map can be drawn as:

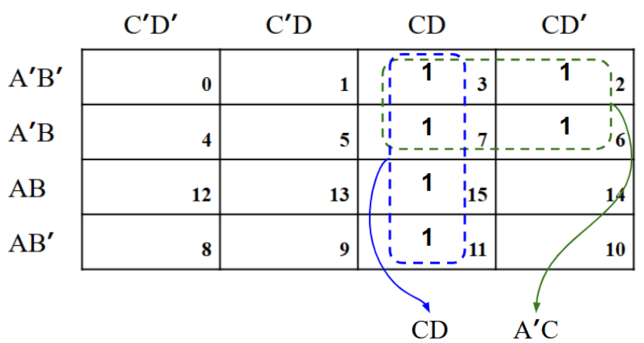


Fig. Solution Q2.98.1

Thus the minimal SOP expression obtained is $\bar{A}C + CD$.

$$\boxed{(B)\ \bar{A}C + CD}$$

Q2.97.1 NAT 1997 1M Ans: 7

The truth table is modified to include A, B, X as inputs.

A	B	X	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

$$Y = \Sigma m(0, 3, 4, 5)$$

The K-Map is drawn as:

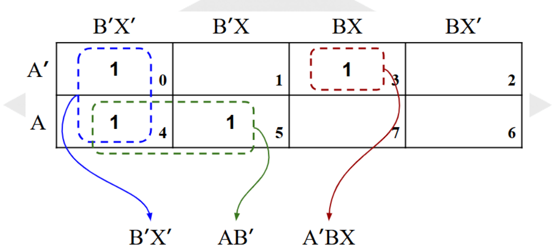


Fig. Solution Q2.97.1

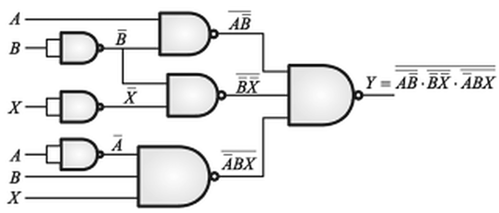
The expression is

$$Y = A\bar{B} + \bar{B}\bar{X} + \bar{A}BX$$

$$Y = \overline{\overline{A\bar{B} + \bar{B}\bar{X} + \bar{A}BX}}$$

$$Y = \overline{\overline{A\bar{B}}.\overline{\bar{B}\bar{X}}.\overline{\bar{A}BX}}$$

Using NAND this circuit can be made as:



Thus we see that 7 NAND gates are required.

$$\boxed{7}$$

Q2.95.1 MCQ 1995 1M Ans: C

From the K-Map, it is observed that no quads or pairs can be formed. Thus the Boolean expression obtained is:

$$Y = (A\bar{B} + \bar{A}B)\bar{C} + (\bar{A}\bar{B} + AB)C$$

$$Y = (A \oplus B)\bar{C} + \overline{(A \oplus B)}C = A \oplus B \oplus C$$

$$\boxed{(C)\ A \oplus B \oplus C}$$

DEBUG INFO

Chapter II

Representation of Boolean Expressions and K-Maps

Total Questions : 18

MCQ : 17

MSQ : 0

NAT : 1

SOLUTIONS

III

Number Systems

Topics

1. Introduction to Number Systems
2. Arithmetic Operations on Numbers
3. Binary Codes and Their Operations
4. Signed Number Representations
5. Complement Arithmetic
6. Floating-Point Representation

Q3.25.1 MCQ 2025 1M Ans: A ● ○ ○

The four-bit binary representation of 5 is:

$$(5)_{10} = (0101)_2$$

Adding the two numbers in 2's complement form:

$$\begin{array}{r} 0101 \\ +\,0101 \\ \hline 1010 \end{array}$$

The carry out of the MSB is discarded in 4-bit arithmetic, leaving the result $(1010)_2$.

Since the most significant bit (MSB) of the result is 1, the number is **negative** in 2's complement representation.

Taking the 2's complement of $(1010)_2$ to find the magnitude:

$$\begin{array}{r} 0101 \\ +\quad 1 \\ \hline 0110 \end{array}$$

$(0110)_2 = (6)_{10}$, so the result is $-(6)_{10}$.

This is an **overflow**: adding two positive numbers (+5 and +5) produces a negative result, which is a classic signed overflow condition in 4-bit arithmetic.

$$\boxed{(A)(-6)_{10}}$$

Key Points

- In n-bit 2's complement arithmetic, the representable range is -2^{n-1} to $2^{n-1} - 1$; for 4-bit, this is -8 to $+7$. Since $5 + 5 = 10 > 7$, overflow occurs.

Mistakes to be avoided

- Do not include the carry-out bit as a 5th bit in the result; 4-bit arithmetic truncates to 4 bits, and the carry is discarded.
- Do not skip checking for overflow. Overflow in 2's complement occurs when two numbers of the *same sign* produce a result of the *opposite sign*. When the result of adding two positive numbers has MSB = 1, overflow has occurred and the result is incorrect as a signed number.

Q3.18.1 MCQ 2018 1M Ans: C ● ○ ○

Converting the integer part $(27)_{10}$ to binary by successive division by 2:

$$\begin{array}{c|ll} 2 & 27 & \\ \hline 2 & 13 & \to 1 \\ \hline 2 & 6 & \to 1 \\ \hline 2 & 3 & \to 0 \\ \hline 2 & 1 & \to 1 \\ \hline & 0 & \to 1 \end{array} \implies (27)_{10} = (11011)_2$$

Converting the fractional part $(0.625)_{10}$ to binary by successive multiplication by 2:

$$\begin{array}{lcl} 0.625 \times 2 = 1.25 & \implies & 1 \\ 0.25 \times 2 = 0.50 & \implies & 0 \\ 0.50 \times 2 = 1.00 & \implies & 1 \end{array} \implies (0.625)_{10} = (0.101)_2$$

Combining both parts:

$$(27.625)_{10} = (11011)_2 + (0.101)_2$$

$$\boxed{(C)\ (11011.101)_2}$$

Key Points

- The integer part is converted by repeated division by 2, reading remainders *upward* (LSB to MSB).
- The fractional part is converted by repeated multiplication by 2, reading the integer parts of the products *downward* (MSB to LSB).
- A terminating decimal fraction does not always terminate in binary; however, $0.625 = 5/8 = 1/2 + 1/8$ terminates exactly in 3 binary places.

Q3.11.1

NAT | 2011 | 1M | Ans: 7

Let the base of the number system be b. Converting each number from base b to base 10:

$$(24)_b = 2 \times b^1 + 4 \times b^0 = 2b + 4$$
$$(14)_b = 1 \times b^1 + 4 \times b^0 = b + 4$$
$$(41)_b = 4 \times b^1 + 1 \times b^0 = 4b + 1$$

Setting up the equation $(24)_b + (14)_b = (41)_b$:

$$(2b + 4) + (b + 4) = 4b + 1$$
$$3b + 8 = 4b + 1$$
$$b = 7$$
$$\boxed{7}$$

Key Points

- Any number $(d_n d_{n-1} \cdots d_1 d_0)_b$ has decimal value $\sum_i d_i \times b^i$; expanding all three numbers in base b and equating gives a linear equation in b.
- The base b must be greater than the largest digit used in any of the numbers; here the largest digit is 4, so $b > 4$. The solution $b = 7$ satisfies this constraint.

Mistakes to be avoided

- Do not treat the given numerals as base-10 numbers and attempt decimal arithmetic; $24 + 14 = 41$ is not true in base 10, confirming the base must be found algebraically.

Q3.09.1

MCQ | 2009 | 1M | Ans: C

The integer part of 1.375 is 1, which is $(1)_2$ directly.
Converting the fractional part $(0.375)_{10}$ to binary by successive multiplication by 2:

$$\begin{array}{lll} 0.375 \times 2 = 0.750 & \rightarrow & 0 \\ 0.75 \times 2 = 1.50 & \rightarrow & 1 \\ 0.50 \times 2 = 1.00 & \rightarrow & 1 \end{array} \implies (0.375)_{10} = (0.011)_2$$

Combining both parts:

$$(1)_{10} + (0.375)_{10} = (1)_2 + (0.011)_2 = (1.011)_2$$
$$\boxed{(C)\ 1.011}$$

Mistakes to be avoided

- Do not stop the multiplication early; continue until the fractional part becomes exactly 0.00 or a repeating pattern is detected.

Q3.08.1 | MCQ | 2008 | 1M | Ans: C

Given:

$$N = (45)_{10} - (45)_{16} \quad \cdots (i)$$

Converting $(45)_{16}$ to decimal:

$$(45)_{16} = 4 \times 16^1 + 5 \times 16^0 = 64 + 5 = 69$$

Substituting into equation (i):

$$N = (45)_{10} - (69)_{10} = (-24)_{10}$$

Finding the 2's complement representation of -24 in 6-bit form.
First, represent $+24$ in binary:

$$+24 \longrightarrow 011000$$

Inverting all bits (1's complement):

$$011000 \longrightarrow 100111$$

Adding 1 to get 2's complement:

$$100111 + 1 = 101000$$

$$\boxed{(C)\ 101000}$$

Key Points

- Alternatively, 2's complement can be found by keeping all trailing zeros and the rightmost 1 unchanged, then inverting all remaining bits to the left.
- In 2's complement, the MSB is the sign bit: 0 for positive, 1 for negative.

Q3.06.1 | MCQ | 2006 | 2M | Ans: D

The 4-bit 2's complement representation of N is:

$$N = \boxed{a_3 \mid a_2 \mid a_1 \mid a_0}$$

In 2's complement, the MSB (sign bit) can be replicated (sign-extended) any number of times without changing the value. Extending N to 6 bits:

$$N = a_3\ a_3\ a_3\ a_2\ a_1\ a_0$$

The 6-bit result after the operations is given as:

a_3	a_3	a_2	a_1	a_0	1

Shifting N left by one bit (equivalent to multiplying by 2) appends a 0 at the LSB:

$$2N = a_3\ a_3\ a_3\ a_2\ a_1\ a_0\ 0$$

Truncating to 6 bits:

$$2N = a_3\ a_3\ a_2\ a_1\ a_0\ 0$$

Adding binary 1 to $2N$:

$$2N + 1 = a_3\ a_3\ a_2\ a_1\ a_0\ 0 + 1 = a_3\ a_3\ a_2\ a_1\ a_0\ 1$$

This matches the given 6-bit result exactly.

$$\boxed{(D) \quad 2N + 1}$$

Key Points

- **Sign extension** in 2's complement: the MSB (sign bit) can be replicated to the left any number of times without changing the numerical value of the number.
- A **left shift by** n **bits** in 2's complement is equivalent to multiplication by 2^n; a right shift by n bits is equivalent to division by 2^n.

Mistakes to be avoided

- Do not forget that a left shift appends a 0 at the LSB, not a 1; the resulting LSB of $2N$ is always 0, making $2N$ even.

Q3.05.1 MCQ 2005 1M Ans: B

Converting the hexadecimal number $(FA)_{16}$ to binary:

$$(FA)_{16} = (1111\ 1010)_2$$

Since the MSB is 1, the number is **negative** in 2's complement representation.
To find the magnitude, take the 2's complement of $(11111010)_2$:
Using the short trick — move from right to left, keep bits up to and including the first '1' unchanged, then invert all remaining bits to the left:

$$11111010 \xrightarrow{\text{2's complement}} 00000110$$

Converting $(00000110)_2$ to decimal:

$$(00000110)_2 = 4 + 2 = 6$$

Therefore the actual value is −6.

$$\boxed{(B)\ \ -6}$$

Key Points

- **Short trick for 2's complement:** scan from LSB to MSB, copy all bits up to and including the first '1', then invert all remaining bits. This avoids the two-step invert-then-add process.
- Hexadecimal to binary conversion: each hex digit maps to exactly 4 bits. F = 1111, A = 1010.
- The 8-bit 2's complement range is −128 to +127; $(FA)_{16} = -6$ lies well within this range.

Mistakes to be avoided

- Do not interpret $(FA)_{16}$ directly as a positive decimal number 250; the MSB being 1 in a signed 2's complement representation always means the number is negative.

Q3.04.1 MCQ 2004 1M Ans: A

Converting the given hexadecimal numbers to binary and performing XOR:

$$7EH \rightarrow 0111\ 1110$$

$$5FH \rightarrow 0101\ 1111$$

$$\text{XOR result:}\quad 0010\ 0001 \implies (21)_{16}$$

Converting $(21)_{16}$ and $(10)_{16}$ to decimal:

$$(21)_{16} = 2 \times 16^1 + 1 \times 16^0 = 32 + 1 = (33)_{10}$$

$$(10)_{16} = 1 \times 16^1 + 0 \times 16^0 = (16)_{10}$$

Computing the product:

$$(21)_{16} \times (10)_{16} = (33)_{10} \times (16)_{10} = (528)_{10}$$

Converting $(528)_{10}$ to hexadecimal by successive division by 16:

$$\begin{array}{c|lcl} 16 & 528 & & \\ \hline 16 & 33 & \rightarrow & 0 \\ \hline & 2 & \rightarrow & 1 \end{array} \implies (528)_{10} = (210)_{16}$$

$$\boxed{(A)\ 0210\ \text{H}}$$

Key Points

- XOR is performed bitwise; each hex digit is first expanded to 4 bits, the XOR is applied bit by bit, and the result is regrouped into hex digits.

Q3.03.1 MCQ 2003 2M Ans: A

Method 1

Interpreting the 4-bit 2's complement numbers:

$$1011 \rightarrow \text{2's complement representation of } -5$$

$$0110 \rightarrow \text{2's complement representation of } +6$$

Adding algebraically:

$$-5 + 6 = +1$$

Representing +1 in 4-bit 2's complement:

$$+1 = 0001$$

Method 2

Performing direct binary addition:

$$\begin{array}{r} 1011 \\ +\,0110 \\ \hline \mathbf{1}0001 \end{array}$$

The carry out of the MSB is discarded in 4-bit 2's complement arithmetic, leaving the result 0001.

$$\boxed{(A)\ 0001}$$

Key Points

- In 2's complement addition, the carry out of the MSB is simply *discarded*; it does not indicate an error when the two operands have *opposite* signs.
- Overflow occurs only when two numbers of the *same* sign produce a result of the opposite sign; here −5 and +6 have opposite signs, so no overflow occurs.
- The 4-bit 2's complement range is −8 to +7; the result +1 lies within this range, confirming no overflow.

Mistakes to be avoided

- Do not include the carry-out bit as a 5th bit in the result; in fixed-width 2's complement arithmetic, the word length is always preserved and the carry is discarded.
- Do not confuse carry-out with overflow; a carry out of the MSB does *not* necessarily mean overflow - check the sign

of both operands and the result to determine overflow.

Q3.99.1

MCQ | 1999 | 2M | Ans: C

The 4-bit 2's complement representation of N is:

$$N = X_3\ X_2\ X_1\ X_0$$

In 2's complement, the MSB (sign bit) can be replicated any number of times to the left without changing the value of the number – this is known as **sign extension**.

To extend from 4 bits to 8 bits, the MSB X_3 is copied 4 additional times:

$$N_{8\text{-bit}} = \boxed{X_3}\boxed{X_3}\boxed{X_3}\boxed{X_3}\boxed{X_3}\boxed{X_2}\boxed{X_1}\boxed{X_0}$$

Examples verifying sign extension:

- 1011 represents −5 in 4-bit; 11111011 represents −5 in 8-bit.
- 0011 represents +3 in 4-bit; 00000011 represents +3 in 8-bit.

$$\boxed{(C)\ X_3\ X_3\ X_3\ X_3\ X_3\ X_2\ X_1\ X_0}$$

Key Points

- Sign extension works for both positive numbers (MSB = 0, padded with zeros) and negative numbers (MSB = 1, padded with ones).

DEBUG INFO

Chapter III

Number Systems

Total Questions : 10

MCQ : 9

MSQ : 0

NAT : 1

SOLUTIONS

IV

Combinational Circuits

Topics

1. Introduction to Combinational Circuits
2. Code Converters and 7-Segment Display
3. Parity Generator and Checker
4. Magnitude Comparators
5. Multiplexers (MUX) and Demultiplexers (DEMUX)
6. Implementing Logical Expressions Using MUX
7. Decoders, Encoders and Priority Encoders
8. Half Adder and Full Adder
9. Delay-Based Problems in Adders
10. Half Subtractor and Full Subtractor
11. Binary Parallel Adder with Delay-Based Problems
12. Look-Ahead Carry Adder
13. Binary Parallel Subtractor
14. Adder and Subtractor Circuits of Different Binary Codes
15. Binary Multiplier

Q4.25.1 MCQ 2025 2M Ans: A

The given MUX circuit is shown below:

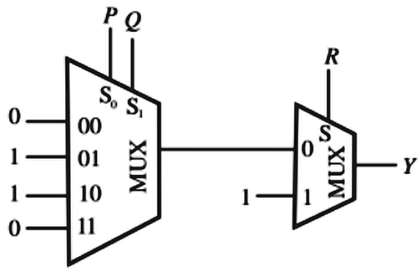


Fig. Solution Q4.25.1

Output of 4:1 MUX is:

$$M_1 = 0(\bar{P}\bar{Q}) + 1(\bar{P}Q) + 1(P\bar{Q}) + 0(PQ) = P \oplus Q$$

Final output is:

$$Y = \bar{R}M_1 + R = R + M_1 = R + (P \oplus Q)$$

$$\boxed{(A)\ R + (P \oplus Q)}$$

Q4.22.1 MSQ 2022 2M Ans: B , D

Output of MUX-1 :

$$M_1 = I_0\bar{S}_0 + I_1S_0$$

For first MUX, $S_0 = B$ and inputs $I_0 = 1, I_1 = 0$. So,

$$M_1 = \bar{B}$$

Output of MUX-2:

$$F = M_2 = I_0\bar{S}_0 + I_1S_0$$

For second MUX, $S_0 = A$ and inputs $I_0 = 1, I_1 = M_1$. So,

$$F = \bar{A} + M_1A = \bar{A} + M_1 = \bar{A} + \bar{B}$$

By De-Morgan's Law,

$$F = \bar{A} + \bar{B} = \overline{A.B}$$

$$\boxed{(B)\ \bar{A} + \bar{B}}\boxed{(D)\ \overline{A.B}}$$

Key Points

- This is a good example of recognizing that $\bar{A} + \bar{B}$ and $\overline{A.B}$ are the same expression due to De Morgan's Law, which is why both options B and D are correct.

Q4.21.1 MCQ 2021 1M Ans: C

The 4x1 MUX has input lines:

$$I_0 = \bar{z} + w \quad , \quad I_1 = 0 \quad , \quad I_2 = zw \quad , \quad I_3 = z\bar{w}$$

Selection lines are

$$S_1 = x \quad , \quad S_0 = y$$

Thus final output equation is:

$$F = \bar{S}_1\bar{S}_0I_0 + \bar{S}_1S_0I_1 + S_1\bar{S}_0I_2 + S_1S_0I_3$$

$$F = \bar{x}\bar{y}(\bar{z}+w) + \bar{x}y(0) + x\bar{y}(zw) + xy(z\bar{w})$$

$$F = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}w + x\bar{y}zw + xyz\bar{w}$$

$$F = \bar{x}\bar{y}\bar{z}\bar{w} + \bar{x}\bar{y}\bar{z}w + \bar{x}\bar{y}\bar{z}w + \bar{x}\bar{y}zw + x\bar{y}zw + xyz\bar{w}$$

$$F = \bar{x}\bar{y}\bar{z}\bar{w} + \bar{x}\bar{y}\bar{z}w + \bar{x}\bar{y}zw + x\bar{y}zw + xyz\bar{w}$$

Writing as minterms indices,

$$F = \Sigma m(0,1,3,11,14)$$

$$\boxed{(C)\ F(x,y,z,w) = \Sigma m(0,1,3,11,14)}$$

Mistakes to be avoided

- Here , $\bar{x}\bar{y}\bar{z}$ appears in two different expanded groups (from $\bar{x}\bar{y}\bar{z}$ and from $\bar{x}\bar{y}\bar{z}w$ before it's absorbed) — always double check that you haven't double counted or lost a minterm when converting SOP with multi-literal terms into a minterm list.

Q4.21.2 | MCQ | 2021 | 1M | Ans: B

From the given circuit diagram, inputs are

$$I_0 = 0 \quad , \quad I_1 = 1 \quad , \quad I_2 = 1 \quad , \quad I_3 = 0$$

Output of given 4x1 MUX is:

$$O = \bar{S}_1\bar{S}_0I_0 + \bar{S}_1S_0I_1 + S_1\bar{S}_0I_2 + S_1S_0I_3 = \bar{S}_1S_0 + S_1\bar{S}_0 = S_1 \oplus S_0$$

$$\boxed{(B)\ O = S_1 \oplus S_0}$$

Key Points

- A multiplexer with n-data select inputs can implement any function of n-variables and some functions of (n + 1) variables. So 2-input XOR can be realised using 4x1 MUX, which employes 2 select lines.

Q4.19.1 | MCQ | 2019 | 1M | Ans: D

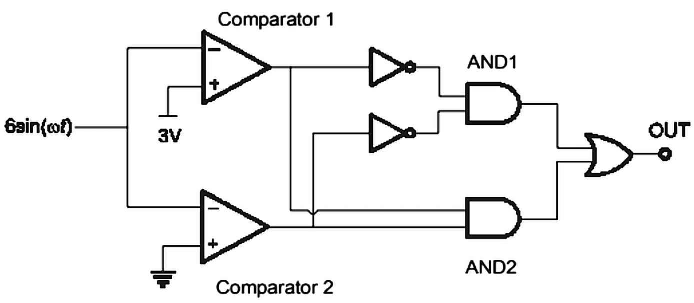
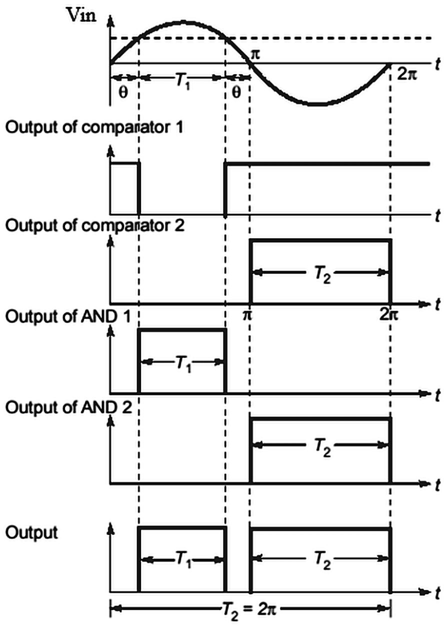


Fig. Solution Q4.19.1

For the given circuit, the transitions can be shown as:



At $\frac{\pi}{6}$ and $\frac{5\pi}{6}$, $V_{in} = 3$, and at π , $V_{in} = 0$. $\therefore$ output is high in range from $\left(\frac{\pi}{6} \text{ to } \frac{5\pi}{6}\right)$ and (π to 2π)

$$= \frac{5\pi}{6} - \frac{\pi}{6} + 2\pi - \pi = \frac{10\pi}{6}$$

$\therefore$ output is high for $\frac{10\pi}{6 \times 2\pi} = \frac{5}{6}$.

$$\boxed{(D)\ \frac{5}{6}}$$

Mistakes to be avoided

- Always sketch $V_{in}(t)$ first and mark threshold crossings before writing X and Y logic, otherwise it's easy to misidentify the intervals where output is high.

Q4.19.2 NAT 2019 1M Ans: 4 ● ● ○

A and B are available at $t = 0$ and stable throughout, therefore outputs of XOR and AND gates will also be stable throughout, and the delay of these gates will not be considered. The maximum time taken for C_i to produce steady state output C_{i+1} is $= t_{AND} + t_{XOR} = 2ns + 2ns = 4ns$.

$$\boxed{4}$$

Key Points

- In ripple-carry-style delay problems, only count delays along the critical path (here AND + XOR), and ignore inputs that are stable from $t = 0$.

Q4.16.1 MCQ 2016 1M Ans: A

From the given circuit , inputs are

$$I_0 = X \quad , \quad I_1 = 0 \quad , \quad I_2 = \bar{X} \quad , \quad I_3 = 1$$

Selection lines are

$$S_1 = Y \quad , \quad S_0 = Z$$

Thus final output equation is:

$$F = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$$

$$F = \bar{Y}\bar{Z}X + Y\bar{Z}\bar{X} + YZ = X\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + \bar{X}YZ + XYZ$$

Writing as minterms indices,

$$F = \Sigma m(2, 3, 4, 7)$$

$$\boxed{(A)\ \Sigma m(2, 3, 4, 7)}$$

Q4.12.1 MCQ 2012 1M Ans: B

Let 2-bit input $A = A_1A_0$ and $B = B_1B_0$. The output is 1 when $A > B$. The truth table is:

$A = A_1A_0$		$B = B_1B_0$		Output
A_1	A_0	B_1	B_0	$Y = A > B$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

It is seen that $Y = 1$ output occurs for 6 combinations of AB .

$$\boxed{(B)\ 6}$$

Mistakes to be avoided

- Always enumerate with truth table or use the known formula $2^{n-1}(2^n - 1)$ pattern logic) to avoid undercounting/overcounting combinations.

Q4.08.1 MCQ 2008 1M Ans: A

From the given circuit , inputs are

$$I_0 = R \quad , \quad I_1 = \bar{R} \quad , \quad I_2 = \bar{R} \quad , \quad I_3 = R$$

Selection lines are

$$S_1 = P \quad , \quad S_0 = Q$$

Thus final output equation is:

$$F = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$$

$$F = \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + PQR$$

$$F = R\overline{(P \oplus Q)} + \bar{R}(P \oplus Q) = P \oplus Q \oplus R$$

$$\boxed{(A)\ P \oplus Q \oplus R}$$

Q4.07.1 MCQ 2007 1M Ans: D

Output of MUX-1 is

$$F_0 = \bar{Y}\bar{X} + YX$$

Output of MUX-1 is

$$F_1 = \bar{F}_0\bar{Z} + F_0 Z = \overline{\bar{X}\bar{Y} + XY}\bar{Z} + (\bar{X}\bar{Y} + XY)Z$$

$$F_1 = (\overline{\bar{X}\bar{Y}}.\overline{XY})\bar{Z} + \bar{X}\bar{Y}Z + XYZ$$

$$F_1 = ((X + Y)(\bar{X} + \bar{Y}))\bar{Z} + \bar{X}\bar{Y}Z + XYZ$$

$$F_1 = \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + \bar{X}\bar{Y}Z + XYZ$$

$$F_1 = (X \oplus Y)\bar{Z} + \overline{(X \oplus Y)}Z = (X \oplus Y) \oplus Z$$

$$\boxed{(D)\ (X \oplus Y) \oplus Z}$$

Key Points

- Cascaded 2:1 MUXes with complementary inputs is a classic way to build XOR chains.

Q4.06.1 MCQ 2006 1M Ans: C

From the given circuit , inputs are

$$I_0 = 1 \quad , \quad I_1 = 1 \quad , \quad I_2 = 1 \quad , \quad I_3 = 0$$

$$I_4 = 0 \quad , \quad I_5 = 0 \quad , \quad I_6 = 1 \quad , \quad I_7 = 0$$

Selection lines are

$$S_2 = A \quad , \quad S_1 = B \quad , \quad S_0 = C$$

The MUX output Y is:

$$Y = \bar{S}_2\bar{S}_1\bar{S}_0 I_0 + \bar{S}_2\bar{S}_1 S_0 I_1 + \bar{S}_2 S_1\bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 + S_2\bar{S}_1\bar{S}_0 I_4 + S_2\bar{S}_1 S_0 I_5 + S_2 S_1\bar{S}_0 I_6 + S_2 S_1 S_0 I_7$$

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C}$$

Output Z is:

$$Z = Y + \bar{C} = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C} + \bar{C}$$

$$Z = \bar{C}(1 + \bar{A}\bar{B} + \bar{A}B + AB) + \bar{A}\bar{B}C = \bar{A}\bar{B}C + \bar{C} = \bar{A}\bar{B} + \bar{C}$$

$$\boxed{(C)\ \bar{C} + \bar{A}\bar{B}}$$

Q4.05.1 MCQ 2005 1M Ans: C

Output O_1 is given by:

$$O_1 = \bar{G}_0 I_0 + G_0 I_1$$

Output O_2 is given by:

$$O_2 = \bar{G}_1 I_2 + G_1 I_3$$

Output O is given by:

$$O = \bar{G}_2 O_1 + G_2 O_2 = \bar{G}_2(\bar{G}_0 I_0 + G_0 I_1) + G_2(\bar{G}_1 I_2 + G_1 I_3)$$

$$O = \bar{G}_2 \bar{G}_0 I_0 + \bar{G}_2 G_0 I_1 + G_2 \bar{G}_1 I_2 + G_2 G_1 I_3$$

$$\boxed{(C)\ \bar{G}_0 \bar{G}_2 I_0 + G_0 \bar{G}_2 I_1 + G_2 \bar{G}_1 I_2 + G_2 G_1 I_3}$$

Q4.04.1 MCQ 2004 1M Ans: C

Given circuit is:

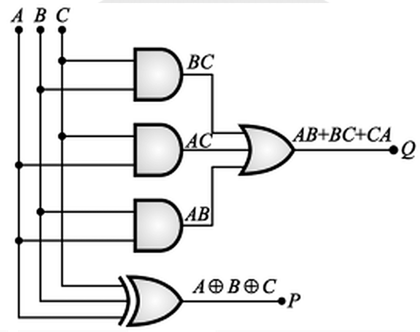


Fig. Solution Q4.04.1

The output is:

$$Q = AB + BC + CA = \text{ Carry of full adder}$$

$$P = A \oplus B \oplus C = \text{ Sum of full adder}$$

$$\boxed{(C)\ \text{Full adder where } P \text{ is the sum and } Q \text{ is the carry}}$$

Key Points

- Recognizing $AB + BC + CA$ and $A \oplus B \oplus C$ as the carry and sum of a full adder respectively is a pattern worth memorizing since it appears frequently.

Q4.03.1 MCQ 2003 1M Ans: C

Given circuit is:

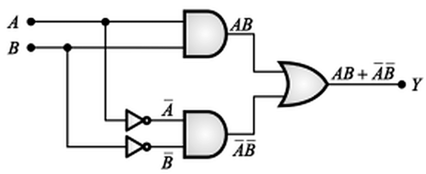


Fig. Solution Q4.03.1

The output is:

$$Y = AB + \bar{A}\bar{B} = A \odot B$$

Output is 1 only if both inputs are equal (property OF XNOR gate) , so this circuit acts as an equality detector.

$$\boxed{(C)\text{ Equality detector}}$$

Key Points

- XNOR works as an equality detector.

Q4.03.2 | MCQ | 2003 | 1M | Ans: C | ● ○ ○

Output V_0 of the multiplexer is:

$$V_0 = \bar{S}.1 + S.0 = \bar{S}$$

From figure, $S = V_0$. Also, $V_{0+} = \bar{V}_0$, i.e., output is toggling. The waveform of the output is :

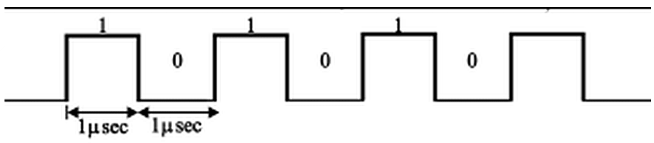


Fig. Solution Q4.03.2

$$\text{Time period } T = 2\ \mu\text{ s}$$
$$\text{Frequency } f = \frac{1}{T} = 0.5\text{ MHz}$$

$$\boxed{(C)\text{ pulse train of frequency 0.5 MHz}}$$

Key Points

- When a MUX's select line is fed by its own output (or a derived signal), look for toggling/oscillatory behavior rather than treating it as static combinational logic — find the time period directly from the timing diagram.

Q4.95.1 | MCQ | 1995 | 2M | Ans: B | ● ○ ○

From the given circuit , inputs are

$$D_0 = 1 \quad , \quad D_1 = 1 \quad , \quad D_2 = 1 \quad , \quad D_3 = C$$

Selection lines are

$$S_1 = A \quad , \quad S_0 = B$$

Thus final output equation is:

$$Q = \bar{S}_1\bar{S}_0D_0 + \bar{S}_1S_0D_1 + S_1\bar{S}_0D_2 + S_1S_0D_3$$
$$Q = \bar{A}\bar{B}C + \bar{A}B + A\bar{B} + AB = \bar{A}(\bar{B}C + B) + A(\bar{B} + B)$$
$$Q = \bar{A}(B + C) + A = A + B + C$$
$$\boxed{(B)\ A + B + C}$$

DEBUG INFO

Chapter IV

Combinational Circuits

Total Questions : 16

MCQ : 14

MSQ : 1

NAT : 1

SOLUTIONS

V

Sequential Circuits

Topics

1. SR Latches
2. JK, D and T Latches and Their Drawbacks
3. Gated Latches
4. Flip-Flops — Introduction and Types
5. Flip-Flop Conversions
6. Race Around Condition and Master-Slave Configuration
7. Binary Asynchronous Counter
8. Non-Binary Asynchronous Counters — Up and Down
9. Decoding the Output of Asynchronous Counters
10. Synchronous Counters
11. Propagation Delay, Setup and Hold Time, Critical Path Delay
12. Shift Registers and Their Applications
13. Ring Counter and Johnson Counter
14. Cascading Counters
15. Delay in Synchronous Counters, Synchronous Clear and Preset Input
16. FSM and Sequence Detector: Introduction to FSM and Design
17. Sequence Generation and Detection
18. Overlapping and Non-Overlapping Sequence Detectors
19. Static Time Analysis: Setup Time, Hold Time and Clock Constraints
20. Critical Path Analysis
21. Timing Violations and Fixes

Q5.26.1 MCQ 2026 2M Ans: C

From the circuit , $D_0 = Q_1$, $D_1 = Q_2$, $D_2 = Q_3$, and the input to Q_3 is XOR of Q_1 and Q_0.

$$D_3 = Q_1 \oplus Q_0$$

Initially, $Q_3Q_2Q_1Q_0 = 1011$, so , $Q_1 = 1$ and $Q_0 = 1$, and $D_3 = 0$.
Thus after first rising edge, $Q_3Q_2Q_1Q_0 = D_3D_2D_1D_0 = 0101$
Now use this new state for the second rising edge: $Q_1 = 0$, $Q_1 = 1$, $D_3 = 0 \oplus 1 = 1$
Also, $D_0 = Q_1 = 0$, $D_1 = Q_2 = 1$, $D_2 = Q_3 = 0$
Hence after second rising edge, $Q_3Q_2Q_1Q_0 = D_3D_2D_1D_0 = 1010$

$$\boxed{(C)\ 1010}$$

Q5.25.1 MCQ 2025 2M Ans: A

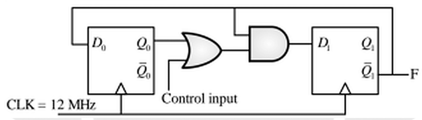


Fig. Solution Q5.25.1

Output $F = \overline{Q_1}$
Characteristic equations of both Flip-Flops are:

$$Q_0(n+1) = D_0 = \overline{Q_1(n)}$$

$$Q_1(n+1) = [Q_0(n) + \text{Control Input}]\ .\ \overline{Q_1(n)}$$

Case I : Control Input = 0

$$Q_0(n+1) = \overline{Q_1(n)}$$
$$Q_1(n+1) = Q_0(n).\overline{Q_1(n)}$$

Q_1	Q_0
0	0
0	1
1	1
0	0

At Q_1 , $T_{ON} = 1T_{clk}$ and $T_{OFF} = 2T_{clk}$,∴ time period of Q_1 : $T_{Q_1} = 3T_{clk}$. This gives, frequency of Q_1 : $f_{Q_1} = \frac{f_{clk}}{3} = \frac{12}{3} = 4MHz$.

$$f_{Q_1} = f_{\overline{Q}_1} = f_F = 4 \text{ MHz}$$

Case II : Control Input = 1

$$Q_0(n+1) = \overline{Q_1(n)}$$
$$Q_1(n+1) = \overline{Q_1(n)}$$

Q_1	Q_0
0	0
1	1
0	0

At Q_1, $T_{ON} = 1T_{clk}$ and $T_{OFF} = 1T_{clk}$, $\therefore$ time period of Q_1 : $T_{Q_1} = 2T_{clk}$. This gives, frequency of Q_1 : $f_{Q_1} = \frac{f_{clk}}{2} = \frac{12}{2} = 6MHz$.

$$f_{Q_1} = f_{\overline{Q}_1} = f_F = 6 \text{ MHz}$$

$$\boxed{(A)\ 4 \text{ MHz and } 6 \text{ MHz}}$$

Mistakes to be avoided

- For a flip-flop with feedback that creates an asymmetric duty cycle (different T_{ON} and T_{OFF}), the output frequency is $\frac{f_{clk}}{\text{total states in cycle}}$, not simply $\frac{f_{clk}}{2}$. Always trace the full state sequence before computing frequency.

Q5.24.1 MCQ 2024 2M Ans: A ● ○ ○

From the given circuit and signals, the following truth table is obtained:

Signals			Inputs			Outputs			Required Output
CLK	DIN	S	D2	D1	D0	Q2	Q1	Q0	Y = (Q2′ Q1 Q0)′
1	1	1	1	0	0	1	0	0	1
2	1	1	1	1	0	1	1	0	1
3	0	1	0	1	1	0	1	1	**0**
4	1	0	0	1	1	0	1	1	**0**
5	1	1	1	0	1	1	0	1	**1**

It is observed that from 3rd to 5th clock cycle, Y takes values 0,0 and 1.

$$\boxed{(A)\ 001}$$

Q5.23.1 MCQ 2023 1M Ans: C ● ○ ○

For the given circuit, Input to shift register $A = \overline{A}_2 \oplus A_0$

Truth table is:

ClK	$\overline{A}_2 \oplus A_0$	$A_3A_2A_1A_0$	$B_3B_2B_1B_0$
–	1	1 1 0 1	1 0 1 0
1	0	1 1 1 0	1 1 0 1
2	1	0 1 1 1	0 1 1 0
3	0	1 0 1 1	1 0 1 1
4		0 1 0 1	1 1 0 1

After 4 clock pulses, $A = 0101$ and $B = 1101$.

$$\boxed{(C)\ A = 0101, B = 1101}$$

Q5.23.2 | MCQ | 2023 | 1M | Ans: A | ● ○ ○

From the given table,

$$Q(t+1) = Q(t) \oplus \text{Input}$$

Comparing this equation with characteristic equation of T flipflop,

$$Q(t+1) = Q(t) \oplus T$$

Hence, it is a T flipflop.

$$\boxed{(A)\ \text{T flip-flop}}$$

Q5.22.1 | MCQ | 2022 | 1M | Ans: C | ● ○ ○

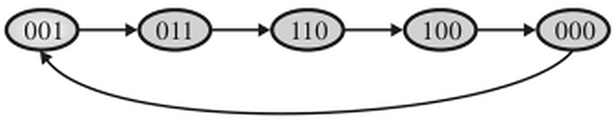
Next state equation of the flip-flop:

$$Q_0^+ = \overline{Q}_1.\overline{Q}_2 \ , \ Q_1^+ = Q_0 \ , \ Q_2^+ = Q_1$$

So state table of given counter is:

Present State			Clock	Next State		
Q_2	Q_1	Q_0		Q_2^+	Q_1^+	Q_0^+
0	0	0	↑	0	0	1
0	0	1	↑	0	1	1
0	1	1	↑	1	1	0
1	1	0	↑	1	0	0
1	0	0	↑	0	0	0

State transition diagram,



It is a MOD-5 counter so frequency is divided by 5.

$$\boxed{(C)\ \text{is a divide-by-5 counter}}$$

Key Points

- Always check whether all states are used or whether the counter resets early.

Q5.21.1 | MCQ | 2021 | 1M | Ans: C | ● ○ ○

Given circuit is synchronous sequential circuit.

Next state equation of Flip-flop A:

$$Q_A^+ = J_A\overline{Q}_A + \overline{K}_A Q_A = \overline{Q}_A(\overline{Q}_A + \overline{Q}_B) + Q_A\overline{(Q_A + Q_B)}$$

$$Q_A^+ = \overline{Q}_A + \overline{Q}_A\overline{Q}_B + \overline{Q}_A\overline{Q}_B Q_A = \overline{Q}_A$$

Next state equation of Flip-flop B:

$$Q_B^+ = T_B \oplus Q_B = Q_A \oplus Q_B$$

State transition table:

Clock	Present State		Next State	
	Q_A	Q_B	Q_A^+	Q_B^+
1	0	0	1	0
2	0	1	1	1
3	1	0	0	1
4	1	1	0	0

State sequence is : $00 \to 10 \to 01 \to 11 \to 00...$

$$\boxed{(C)\ 00 \to 10 \to 01 \to 11 \to 00...}$$

Q5.20.1 NAT 2020 1M Ans: 7

Both flipflop outputs A and B toggle if T_A and T_B are both 1.

$$T_A = \overline{X.B} \qquad T_B = X$$

Initially A=1 , B=1 and x takes values 1 , 0 , 1 in next 3 clock pulses.

Clock	X	A	B	T_A	T_B	A^+	B^+	Y^+
1	1	1	1	0	1	1	0	1
2	0	1	0	1	0	0	0	0
3	1	0	0	1	1	1	1	1

Output $(ABY)_2$ after 3 clocks is $(111)_2 \to$ decimal equivalent is 7 .

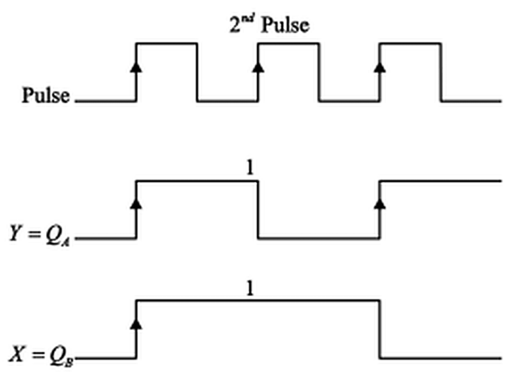
$$\boxed{7}$$

Q5.19.1 MCQ 2019 1M Ans: D

As positive output of FF-1 is connected as positive pulse to FF-2, this configuration appears to be a down-counter. $\therefore$ it is a 2-bit asynchronous counter, which counts,

Pulse	X	Y
1	0	0
2	1	1
3	1	0
4	0	1
1	1	1
2	1	0

Timing diagram of signals:



It is observed that value of X and Y just before arrival of second pulse is $X = 1, Y = 1$.

$$\boxed{(D)\ X = 1, Y = 1}$$

Q5.18.1 MCQ 2018 1M Ans: C

From figure, inputs are"

$$J_1 = Q_1 + Q_2 \quad , \quad K_1 = \bar{Q}_1 + \bar{Q}_2 \quad , \quad J_2 = \bar{Q}_1 + Q_2 \quad , \quad K_2 = Q_1 + \bar{Q}_2$$

Next state equation/characteristic equation of J-K flipflop is given by

$$Q^+ = J\bar{Q} + \bar{K}Q$$

Thus the state table is made until a state is repeated:

Present State		Inputs				Next State	
Q_1	Q_2	J_1	K_1	J_2	K_2	Q_1^+	Q_2^+
0	0	0	1	1	1	0	1
0	1	1	1	1	0	1	1
1	1	1	0	1	1	1	0
1	0	1	1	0	1	0	0
0	0	0	1	1	1	0	1

Output sequence of Q_1Q_2 is 00 , 01 , 11 , 10 , 00 ...

$$\boxed{(C)\ 00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00...}$$

Q5.18.2 NAT 2018 1M Ans: 10

Given : 3 digit counter with starting state as 000.

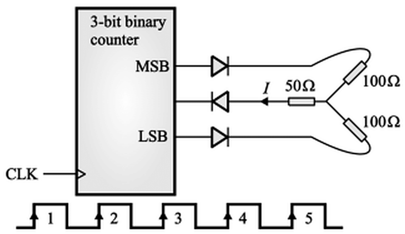
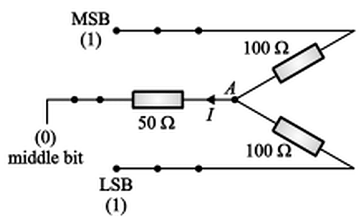
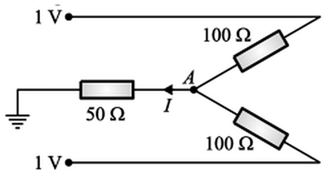


Fig. Solution Q5.18.2

For 5^{th} clock cycle, the state of the counter is 101, and all the diodes are short-circuited after that. The equivalent figure is:



It is given that Logic 1 $\implies$ 1V , and Logic 0 $\implies$ Ground. So this circuit can be modified as:



By applying KCL at node A :

$$\frac{V_A}{50} + \frac{V_A - 1}{100} + \frac{V_A - 1}{100} = 0$$

$$2V_A + V_A - 1 + V_A - 1 = 0$$

$$V_A = 0.5V$$

$$I = \frac{V_A}{50} = 10 \text{ mA}$$

$$\boxed{10}$$

Q5.17.1 | MCQ | 2017 | 1M | Ans: B

The given circuit may be modified as:

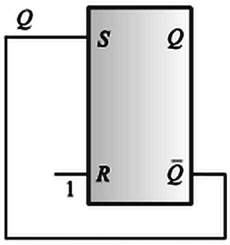


Fig. Solution Q5.17.1

The output of SR latch is given by $Q^+ = S + \bar{Q}R = Q + \bar{Q}.1 = 1$ So $Q^+\bar{Q}^+ = 10$

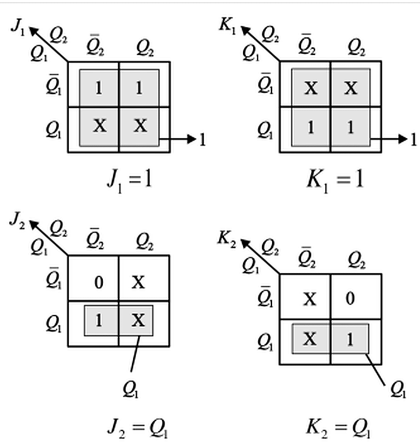
$$\boxed{(B)\ 10}$$

Q5.16.1 | MCQ | 2016 | 1M | Ans: B

Given circuit is 2-bit synchronous sequential circuit. Circuit operated on negative edge clock. Count sequence is given as $00 \to 10 \to 01 \to 11 \to 00 \ldots$. From the given sequence , state table is:

Present State		Next State		Inputs			
Q_1	Q_2	Q_1^+	Q_2^+	J_1	K_1	J_2	K_2
0	0	1	0	1	X	0	X
0	1	1	1	1	X	X	0
1	0	0	1	X	1	1	X
1	1	0	0	X	1	X	1

The K-maps for J_1 , K_1 , J_2 , K_2 are:



From the above K-maps,

$$J_1 = 1\ ,\ J_2 = Q_1\ ,\ K_1 = 1\ ,\ K_2 = Q_1$$

$$\boxed{(B)\ J_1 = 1\ ,\ K_1 = 1\ ,\ J_2 = Q_1\ ,\ K_2 = Q_1}$$

Q5.15.1 MCQ 2015 1M Ans: B

Given clock frequency = f MHz

For N-decade counter, maximum pulse width = $\dfrac{10^N}{f_{clock}} = \dfrac{10^N}{f}$

Resolution = 1 count = 1 $.T_{clock} = \dfrac{10^N}{f}$

Range of pulse width is given by 0 to $\dfrac{(10^N - 1)}{f}$.

$$\boxed{(B)\ \frac{1}{f}\ \textbf{and}\ \frac{(10^N - 1)}{f}}$$

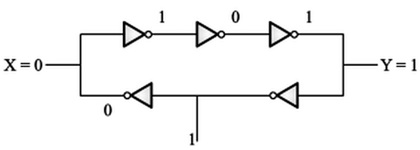
Key Points

- Resolution equals one clock period; range is determined by counter capacity ($10^N - 1$ counts), not by frequency alone

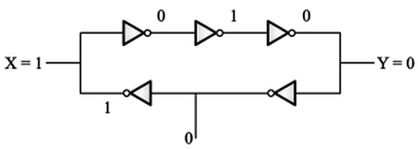
Q5.15.2 MCQ 2015 1M Ans: D

Given: all gates have non-zero delay. The given circuit is an astable multivibrator circuit, with odd number of inverters in the loop. In such a circuit, irrespective of the position of the output, it always toggles.

When switch is closed,



When switch is open,



$\boxed{(D)\text{ X and Y both toggle continuously}}$

Key Points

- Any ring of an odd number of inverters (with non-zero gate delay) is inherently unstable and oscillates — a classic ring-oscillator recognition pattern.

Q5.14.1 MCQ 2014 1M Ans: C ● ○ ○

Range of frequency = 5 kHz to 10 kHz
Resolution = 100 Hz
Minimum clock frequency = $10 \times 10^3 \times 100 = 1$ MHz
Minimum number of bits (n) is given by

$$5000 \geq \frac{10^6}{2^n} \implies 2^n \geq 200 \implies n \geq 8$$

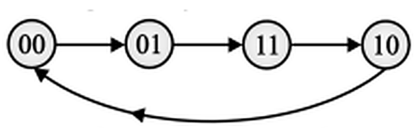
$\boxed{(C)\ 1\text{ MHz}, 8\text{ bits}}$

Q5.13.1 MCQ 2013 1M Ans: B ● ○ ○

From figure, it is a 2-bit synchronous counter since clock input is same for both flipflops. Inputs are : $D_1 = Q_0$, $D_0 = \bar{Q}_1$.
The next state equation/characteristic equation of D flipflop is $Q_{n+1} = D$. So state table is:

Present State		Inputs		Clock	Next State	
Q_1	Q_0	$D_1 = Q_0$	$D_0 = \bar{Q}_1$		Q_1^+	Q_0^+
0	0	0	1	↑	0	1
0	1	1	1	↑	1	1
1	1	1	0	↑	1	0
1	0	0	0	↑	0	0

From this table, the state transition diagram is:



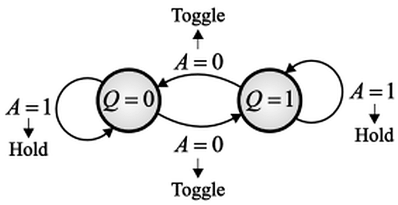
$\boxed{(B)\ 00 \to 01 \to 11 \to 10 \to 00...}$

Q5.12.1 MCQ 2012 2M Ans: D ● ○ ○

Method 1

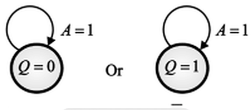
Previous State Q_n	Input A	Next State $Y = D = Q_{n+1}$	State
0	0	1	Toggle
0	1	0	Hold
1	0	0	Toggle
1	1	1	Hold

The truth table corresponds to the given state diagram:

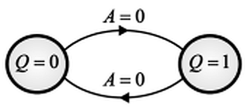


Method 2

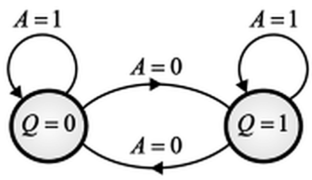
If $A = 1$ then $Y = X_1 = D = Q$. Thus , $Q_{n+1} = Q$.



If $A = 0$ then $Y = X_0 = D = \bar{Q}$. Thus , $Q_{n+1} = \bar{Q}$.



Overall state diagram obtained:



$\boxed{(D)}$

Q5.12.2 MCQ 2012 1M Ans: A ● ○ ○

Race around condition occurs in J-K flip-flop when both inputs are 1 and flipflops are level-triggered. However, given circuit is an S-R flip-flop, where race around condition never occurs.

$\boxed{(A) \text{ does not occur}}$

Mistakes to be avoided

- Race-around condition is specific to level-triggered JK flip-flops with both inputs at 1; SR flip-flops simply don't have this problem because the $S = R = 1$ input combination is invalid/undefined rather than oscillatory. Don't confuse "race-around" with general "invalid state" issues.

Q5.11.1 MCQ 2011 1M Ans: D ● ● ○

When $Q_2Q_1Q_0 = 000$, the output $\bar{Y}_0$ will be active, which will clear the flip-flop and up counting will start:

Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1

When $Q_2Q_1Q_0 = 101$, then $\bar{Y}_5$ will be active , which will preset the flip-flop and down counting will start:

Q_2	Q_1	Q_0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

and then the cycle repeats. So the counter outputs in decimal are 0 , 1 , 2 , 3 , 4 , 5 , 4 , 3 , 2 , 1 , 0 , 1 , 2 .

$$\boxed{(D)\ 0, 1, 2, 3, 4, 5, 4, 3, 2, 1, 0, 1, 2}$$

Q5.09.1 MCQ 2009 1M Ans: C ● ○ ○

From the figure, initially $J = 1, K = \bar{Q}$

The next state equation is $Q^+ = J\bar{Q} + \bar{K}Q$

J	Q	K	Q^+
1	0	1	1
1	1	0	1
1	1	0	1
...	...	...	...
1	1	0	1

The output sequence at Q is 1111...

$$\boxed{(C)\ 1111...}$$

Q5.09.2 MCQ 2009 1M Ans: A ● ○ ○

From figure,

(i) It is a positive edge-triggered flip-flop.
(ii) Normal output of one flip-flop is used as clock input for next flip-flop.
(iii) Output of counter is taken from normal output of final flip-flop.
It is an 8-bit down counter.

$$\boxed{(A)\ \text{always down}}$$

Key Points

- In ripple counters, "normal output drives next clock" + "positive edge-triggered" combination is the standard signature of a down-counter; if instead the complemented output drove the next clock, it would be an up-counter.

Q5.08.1 MCQ | 2008 | 1M | Ans: A

From figure,

$$Y = \overline{\overline{Q_1Q_3}\ \overline{Q_2Q_3}} = Q_1Q_3 + Q_2Q_3 = Q_3(Q_1 + Q_2)$$

$$\boxed{(A)\ Q_3(Q_1 + Q_2)}$$

Q5.08.2 MCQ | 2008 | 1M | Ans: C

From figure, given circuit is reset if $A = 0$, where $A = \bar{Y}$. From previous question,

$$A = \overline{Q_3(Q_1 + Q_2)} = \bar{Q}_3 + \bar{Q}_1\bar{Q}_2$$

In the figure, flops are negative edge triggered and output is taken from normal output, so it is up counter.

Q_3	Q_2	Q_1	Q_0	A
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0

Counter counts from 0000 to 1001. When it reaches state 1010 , clear input is active and all flops are cleared. Therefore it is a Mod-10 counter.

$$\boxed{(C)\ \text{Mod-10 Counter}}$$

Q5.07.1 MCQ | 2007 | 1M | Ans: B

From figure,

Pulse width = T

Time period of clock pulse = T_C

From given figure, $T = 6T_C$, but during this $6T_C$ duration, number of active edges of clocks = 7. So error will be T_C in $7T_C$ duration.

The error in measurement is given by,

$$\text{Error}(x) = \frac{T_C}{7T_C} \times 100$$

$$x = \frac{T_C}{T_C + 6T_C} \times 100 = \frac{T_C}{T_C + T} \times 100$$

Multiplying both denominator and numberator by f_c,

$$x = \frac{T_C.f_c}{(T_C + T).f_c} \times 100 = \frac{1}{Tf_c + 1} \times 100$$

$$\text{i.e. }, x < \frac{100}{T f_c} \implies T < \frac{100}{x f_c}$$

$$\boxed{(B)\ T < \frac{100}{x f_c}}$$

Q5.07.2 MCQ 2007 1M Ans: D

Given circuit is shown below:

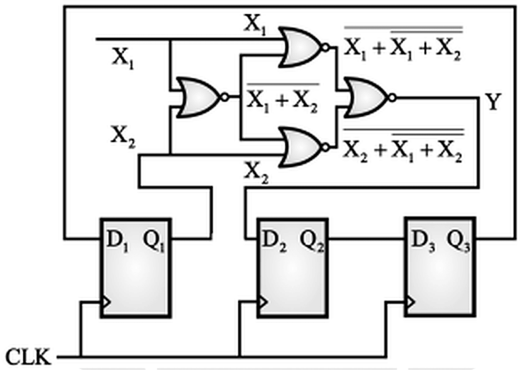


Fig. Solution Q5.07.2

From figure,

$$Y = \overline{\overline{X_1 + \overline{(X_1 + X_2)}} + \overline{\overline{(X_1 + X_2)} + X_2}}$$

$$Y = (X_1 + (\overline{X_1 + X_2}))((\overline{X_1 + X_2}) + X_2) = (X_1 + \bar{X}_1\bar{X}_2)(X_2 + \bar{X}_1\bar{X}_2)$$

$$Y = (X_1 + \bar{X}_2)(X_2 + \bar{X}_1) = \overline{X_1 \oplus X_2}$$

$$\boxed{(D)\ Y = \overline{X_1 \oplus X_2}}$$

Q5.07.3 MCQ 2007 1M Ans: B

For a D flipflop, the next state output is given as $Q_{n+1} = D$.

From figure,

$$D_1 = Q_3$$

$$D_2 = Y = \overline{X_1 \oplus X_2} = \overline{Q_3 \oplus Q_1}$$

$$D_3 = Q_2$$

Truth table for the given counter is shown below:

Q_1	Q_2	Q_3	D_1	D_2	D_3	Q_1^+	Q_2^+	Q_3^+
0	0	0	0	1	0	0	1	0

$\therefore$ the next state after 1 clock pulse is 010.

$$\boxed{(B)\ 010}$$

Q5.07.4 MCQ 2007 1M Ans: C

From figure , for every p_v there are 3 clock pulses. For initial two clock pulses, the counter acts as up counter, and for the 3rd clock pulse, it acts as down counter. Thus for 3 clock pulses, the count is incremented by 1 as:

Clk	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	0	1
4	0	0	1	0
5	0	0	1	1
6	0	0	1	0
...	...	...	...	
42	1	1	1	0
43	1	1	1	1
44	0	0	0	0

Thus the counter reaches 000 after 44th pulse.

$$\boxed{(C)\ \text{44th clock pulse}}$$

Q5.07.5 MCQ 2007 1M Ans: C

From figure

$$T_A = Q_A + Q_B \quad , \quad T_B = \bar{Q}_A + Q_B$$

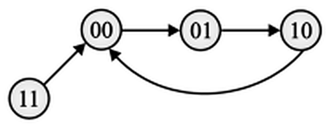
Next state equation of T flipflop is

$$Q^+ = T \oplus Q$$

Truth table for the given circuit is:

Present State		**Inputs**		**Clock**	**Next State**	
Q_A	Q_B	$T_A = Q_A + Q_B$	$T_B = \bar{Q}_A + Q_B$		Q_A^+	Q_B^+
0	0	0	1	↑	0	1
0	1	1	1	↑	1	0
1	0	1	0	↑	0	0
1	1	1	1	↑	0	0

The state transition diagram for output is:



From the given options,

- **Option A:** 01 → 00

  It is observed that state 00 can be reached from state 01 in 2 clock pulses.

- **Option B:** 11 → 01

  It is observed that state 01 can be reached from state 11 in 2 clock pulses.

- **Option C:** $01 \rightarrow 11$

  It is observed that state 11 cannot be reached from state 01.

- **Option D:** $01 \rightarrow 10$

  It is observed that state 10 can be reached from state 01 in 1 clock pulse.

$$\boxed{(C)\ 01 \rightarrow 11}$$

Q5.06.1 MCQ 2006 1M Ans: B ● ○ ○

- When X=0, output $Y^+ = Y$
- When X=1, output $Y^+ = \bar{Y}$

Thus the circuit can be used as a clock generator if X=1.

$$\boxed{(B)\ X = 1}$$

Q5.06.2 MCQ 2006 1M Ans: A ● ○ ○

Given circuit is a 2-bit synchronous counter. Initial state is 00 and second flop works in toggle mode. From figure, inputs are:

$$J_0 = K_0 = \bar{Q}_1, \quad , \quad J_1 = K_1 = 1$$

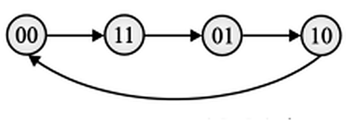
The next state equation of JK flipflop is:

$$Q^+ = J\bar{Q} + \bar{K}Q$$

Truth table is:

Present State		Inputs				Clock	Next State	
Q_1	Q_0	$J_1 = 1$	$K_1 = 1$	$J_0 = \bar{Q}_1$	$K_0 = \bar{Q}_1$		Q_1^+	Q_0^+
0	0	1	1	1	1	↑	1	1
1	1	1	1	0	0	↑	0	1
1	0	1	1	1	1	↑	1	0
1	0	1	1	0	0	↑	0	0

The state transition of the output is:



Thus the count sequence is 00 - 11 - 01 - 10 - 00...

$$\boxed{(A)\ 00 - 11 - 01 - 10 - 00}$$

Q5.05.1 MCQ 2005 1M Ans: D

Given circuit is shown below:

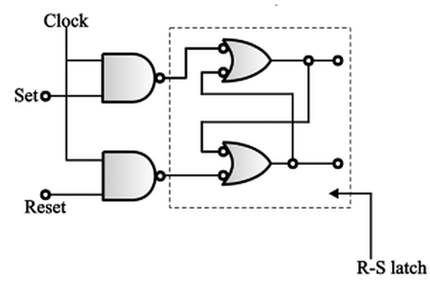


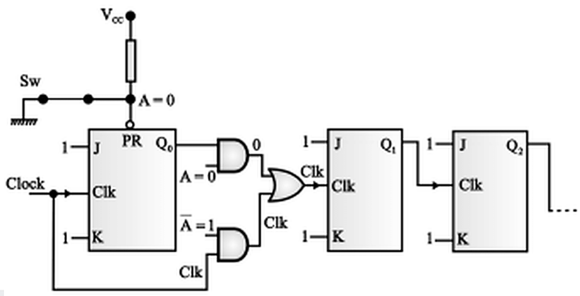
Fig. Solution Q5.05.1

When Clock is high, then NAND gates work as inverters so, the two NAND gates invert the latching action.

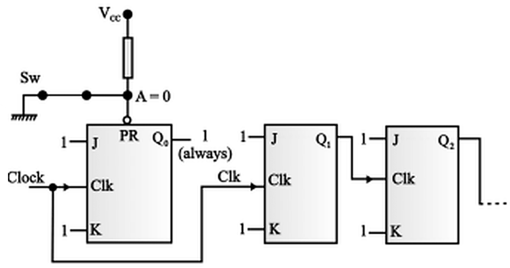
$\boxed{(D) \text{ invert the latching action}}$

Q5.05.2 MCQ 2005 1M Ans: B

When switch is closed, modified circuit is shown below:



The above circuit can be reduced as:



When switch is closed $A = 0$, Preset is active and Q_0 is always 1, i.e., the LSB is 1 so numbers counted by counter is always odd.

$\boxed{(B) \text{ Counter outputs are only odd numbers}}$

Q5.04.1 MCQ 2004 1M Ans: C ● ● ○

Given:

- Delay of AND gate = 5 ns
- Flipflop set time and worst case delay are 5 ns and 15 ns respectively

Since two flip-flops have fixed input , so there is no requirement of setup time.

Total time at the output of AND gate = 15 ns + 15 ns + 5 ns = 35 ns

Also, the output at AND gate is S input of flipflop which must set at least 5 ns. So minimum time to operate circuit faithfully is given by

$$T \geq 35 \text{ ns } + 5 \text{ ns } = 40 \text{ ns}$$

Maximum clock rate for the circuit to operate faithfully is given by:

$$f = \frac{1}{T} = \frac{1}{40 \text{ ns}} = 25 \text{ MHz}$$

$$\boxed{(C)\ 25 \text{ MHz}}$$

Q5.04.2 MCQ 2004 1M Ans: A ● ○ ○

Initially, $Q = A = B = 0$. JK flipflop is operated in toggle mode. Counter is positive edge counter therefore it counts next step when Q changes from 0 to 1, because Q works as a clock. The sequence is given by,

Q	Clock	Counter Clock	A	B
0	↑	1	0	1
1	↑	0	0	1
0	↑	1	1	0

Thus after 3 clock pulses, QAB will be 110.

$$\boxed{(A)\ 110}$$

Q5.03.1 MCQ 2003 1M Ans: B ● ● ○

Given circuit is a 2-bit synchronous counter. From figure,

$$J_0 = \bar{Q}_1 \quad , \quad K_0 = 1$$
$$J_1 = Q_0 \quad , \quad K_1 = 1$$

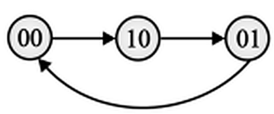
For a JK flipflop , the next state output is given by

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

Truth table for the given counter is:

Present State		Inputs				Clock	Next State	
Q_0	Q_1	$J_0 = \bar{Q}_1$	$K_0 = 1$	$J_1 = Q_0$	$K_1 = 1$		Q_0^+	Q_1^+
0	0	1	1	0	1	↑	1	0
1	0	1	1	1	1	↑	0	1
0	1	0	1	0	1	↑	0	0

The state transition diagram for output is:



$$\boxed{(B)\ 00\ ,\ 10\ ,\ 01\ ,\ 00\ ,\ 10\ ,\ 01\ ,\ ...}$$

Q5.98.1 MCQ 1998 1M Ans: A

The minimum number of flipflops required for mod-N counter is given by

$$2^n \geq N$$

where n is the minimum number of flipflops.

$$2^n \geq 2 \implies n = 1$$

$$\boxed{(A)\ 1}$$

Q5.97.1 MCQ 1997 1M Ans: C

Given:

- 4-bit synchronous counter with a series carry using flipflop and AND gate
- Propagation delays of flipflop and AND gate are 30 ns and 10 ns respectively

Frequency of operation of synchronous counter is given by:

$$f \leq \frac{1}{30 \text{ ns} + 10 \text{ ns}}$$

So the time period between successful clock pulses $T = \frac{1}{f} =] \frac{1}{40 \text{ ns}} = 40 \text{ ns}$

$$\boxed{(C)\ 40 \text{ ns}}$$

Q5.95.1 MCQ 1995 1M Ans: D

From figure,

$$J_0 = K_0 = 0 \quad , \quad J_1 = K_1 = 1$$
$$J_2 = K_2 = Q_1 \quad , \quad J_3 = K_3 = Q_1Q_2$$

For a JK flipflop , the next state output is given by

$$Q^+ = J\bar{Q} + \bar{K}Q$$

$$Q_3^+ = Q_1Q_2\bar{Q}_3 + \overline{Q_1Q_2}Q_3 = Q_1Q_2 \oplus Q_3$$
$$Q_2^+ = Q_1\bar{Q}_2 + \bar{Q}_1Q_2 = Q_1 \oplus Q_2$$
$$Q_1^+ = \bar{Q}_1$$
$$Q_0^+ = Q_0$$

Truth table for the counter is:

Q_3	Q_2	Q_1	Q_0	Clock	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	1	$\uparrow$	0	0	1	1

$$\boxed{(D)\ 0011}$$

Q5.95.2 MCQ 1995 1M Ans: A

The truth table of the given circuit is:

A	B	Q	Q^+	State
0	0	Q	Q	Hold
0	1	1	0	Set
1	0	0	1	Reset
1	1	1	1	Invalid

As clock is not present in the given circuit, it is an R-S latch with $R = \bar{A}$ and $S = \bar{B}$.

$$\boxed{(A)\ \text{R-S latch}}$$

Q5.94.1 MCQ 1994 1M Ans: C

The minimum number of flipflops required for mod-N counter is given by

$$2^n \geq N$$

where n is the minimum number of flipflops.

$$2^n \geq 10 \implies 16 \geq 10 \implies n = 4$$

$$\boxed{(C)\ 4}$$

Q5.94.2 MCQ 1994 1M Ans: D

For a JK flipflop the truth table is:

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

$$\boxed{(D)\ \bar{Q}_n}$$

Q5.93.1 MCQ 1993 1M Ans: A

2-bit synchronous counter , initially holds $Q_1 = 1, Q_2 = 0$ before the arrival of a clock. From figure,

$$J_1 = Q_2 \quad , \quad K_1 = 1 \quad , \quad K_2 = \bar{Q}_1$$

Since the value of J_2 is not given by default, we consider $J_2 = 1$.

The next state equation of JK flipflop is given by

$$Q_n^+ = J\bar{Q}_n + \bar{K}Q_n$$

Truth table for given counter is:

Q_1	Q_2	$J_1 = Q_2$	$K_1 = 1$	$J_2 = 1$	$K_2 = \bar{Q}_1$	Clock	Q_1^+	Q_2^+
1	0	0	1	1	0	$\uparrow$	0	1

Thus , the states after arrival of clock edge are $Q_1 = 0$ and $Q_2 = 1$.

$$\boxed{(A)\ Q_1Q_2 = 01}$$

DEBUG INFO

Chapter V

Sequential Circuits

Total Questions : 43

MCQ : 41

MSQ : 0

NAT : 2

SOLUTIONS

VI

ADC & DAC

Topics

1. Introduction to DAC
   — Weighted Resistor and R-2R Ladder Type
2. Introduction to ADC
   — Counter Type
   — SAR Type ADC
   — Flash Type ADC
   — Single Slope and Dual Slope ADC
   — Sample and Hold Operation in ADC

Q6.26.1 NAT 2026 1M Ans: 63 ● ○ ○

For an n-bit flash ADC, the number of comparators required is

$$N = 2^n - 1$$

For $n = 6$,

$$N = 2^6 - 1 = 64 - 1 = 63$$

$$\boxed{63}$$

Q6.25.1 MCQ 2025 2M Ans: A ● ○ ○

For input 110, switches D_2 and D_1 are closed and D_0 is open, as shown.

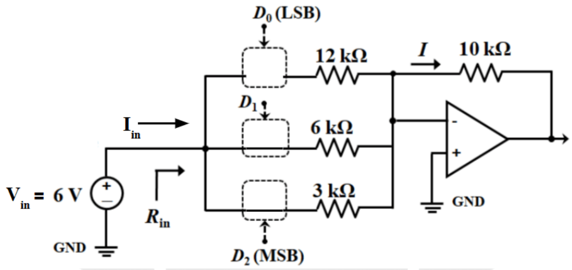


Fig. Solution Q6.25.1

Applying Nodal Analysis at inverting terminal of OPAMP,

$$\frac{0 - V_{in}}{6k} + \frac{0 - V_{in}}{3k} + I_{in} = 0$$

$$I_{in} = \frac{V_{in}}{6k} + \frac{V_{in}}{3k} = \frac{V_{in}}{2k}$$

$$R_{in} = \frac{V_{in}}{I_{in}} = 2k\Omega$$

$$\text{and } I = I_{in} = \frac{V_{in}}{R_{in}} = 3 \text{ mA}$$

$$\boxed{(A)\ R_{in} = 2 \text{ k } \Omega \text{ and } I = 3 \text{ mA}}$$

Q6.25.2 NAT 2025 2M Ans: 125 ● ○ ○

A dual-slope ADC has:

- Fixed integration time = 100 ns
- Reference voltage = -5 V
- Input Voltage = 1.25 V

Dual slope ADC time relation:

$$t_{\text{de-integration}} = \left(\frac{V_{in}}{V_{ref}}\right) T_{\text{integration}}$$

$$t_{\text{de-integration}} = \left(\frac{1.25}{5}\right) 100 = 25 \text{ ms}$$

Total time = $t_{\text{de-integration}} + T_{\text{integration}} = 100 + 25 = 125$ ms

$$\boxed{125}$$

Q6.25.2 | NAT | 2025 | 1M | Ans: 125

Given ADC has:

- Full scale voltage = 1.4 V
- Resolution = 200 mV
- Bandwidth = 500 MHz

$$\text{Resolution} = \frac{V_{\max} - V_{\min}}{2^N - 1}$$

$$2^N - 1 = \frac{1.4}{0.2} = 7$$

$$N = 3 \text{ bits}$$

The ADC samples the data at Nyquist rate:

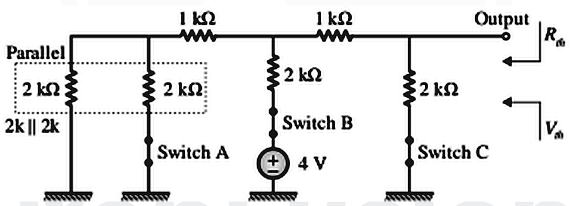
$$f_N = 2f_m = 2 \times 500 \text{ MHz} = 1 \text{ GHz} = 10^9 \text{ samples/sec}$$

Parallel data rate = 3×10^9 bps = 3 Gbps
Serial data rate = 1 Gbps

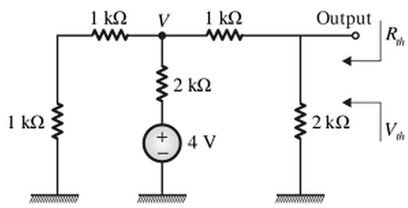
$\boxed{3}$

Q6.22.1 | MCQ | 2022 | 2M | Ans: A

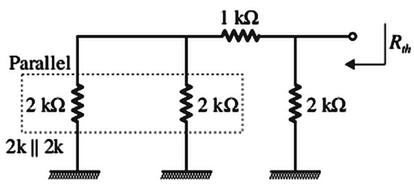
The circuit modified according to opened and closed switches are:



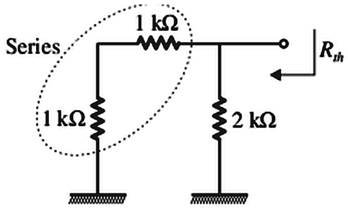
Circuit can be further simplified as:



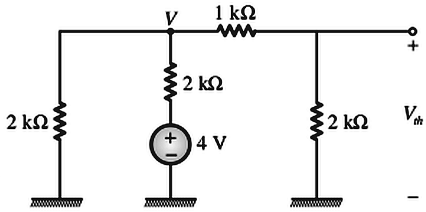
Calculation of R_{th}: Remove all independent sources , i.e. , short circuit the voltage sources and open circuit the current sources. The circuit obtained is:



Circuit may be further simplified as:



Calculation of V_{th}:



Applying KCL at node V:

$$\frac{V-4}{2}+\frac{V}{2}+\frac{V}{3}=0$$

$$3V-12+2V+3V=0$$

$$V=1.5\text{ V}$$

Applying voltage division rule,

$$V_{th}=1.5\times\frac{2}{3}=1\text{ V}$$

$$\boxed{(A)\ V_{th}=1\text{ V}\ ,\ R_{th}=1\text{ k}\Omega}$$

Q6.22.2 | NAT | 2022 | 2M | Ans: 10 | ● ○ ○

Given ADC has:

- Full scale voltage $V_{FS}=10$ V
- Resolution = 0.01 V

$$\text{Resolution}=\frac{\text{Full scale voltage}}{\text{Step size}}$$

$$0.01=\frac{10}{2^N-1}$$

$$2^N=1001\implies n\approx 9.96$$

$$\therefore n=10$$

$$\boxed{10}$$

Q6.21.1 | NAT | 2021 | 1M | Ans: 4.9 to 5.1 | ● ○ ○

Given ADC has:

- Full scale voltage $V_{FS}=10.23$ V
- Number of bits $n=10$
- Number of steps $=2^n-1=1023$

It is known that

$$\text{Full scale voltage} = \text{Resolution} \times \text{Number of steps}$$

$$10.23\text{V} = \text{Resolution} \times 1023$$

$$\text{Resolution} = 10 \text{ mV}$$

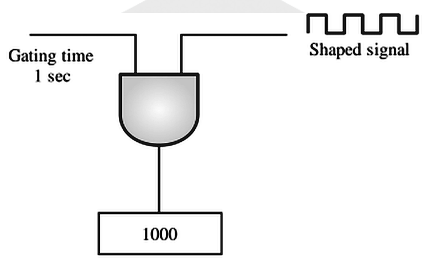
$$\text{Maximum quantization error of ADC} = \frac{\text{Resolution}}{2} = 5 \text{ mV}$$

$\boxed{5}$

Q6.21.2 MCQ 2021 1M Ans: A

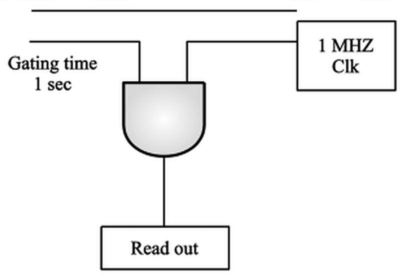
Method 1

Case 1: Frequency mode of operation.



$$f_{\text{signal}} = 1000 \text{ Hz}$$

Case 2: Time period mode of operation.



$$\text{Reading} = \frac{1 \text{ ms}}{1\ \mu\text{s}} = 1000$$

Method 2

Case 1: Frequency mode of operation.

T_s = Gating interval = 1 sec

Reading = 1000

$$\text{Reading of counter} = \text{Gating interval} \times \text{Frequency}$$

$$1000 = 1 \times f$$

$$f = 1000 \text{ Hz}$$

Case 2: Time period mode of operation.

Frequency of pulse passing through gate = f_s = 1 MHz

$$\text{Gating interval} = \frac{1}{f} = \frac{1}{1000} = 1 \text{ ms}$$

$$\text{Reading of counter} = \text{Gating interval} \times \text{Frequency}$$

$$\text{Reading of counter} = 10^{-3} \times 10^{6} = 1000$$

$$\boxed{(A)\ 1000}$$

Q6.21.3 | MSQ | 2021 | 2M | Ans: B , D | ● ● ○

n bit flash ADC requires 2^n-1 comparators. So 4 bit flash ADC requires 15 comparators.

Also a change in input by $\frac{V}{2^n}$ flips LSB of the output of an n bit flash ADC. $\therefore$ a change in input voltage by $\frac{V}{16}$ will always flip LSB of the output.

$$\boxed{(B)\ \text{A change in input voltage by } \frac{V}{16} \text{ will always flip LSB of the output.}}$$

$$\boxed{(D)\ \text{The ADC requires 15 comparators.}}$$

Q6.20.1 | NAT | 2020 | 2M | Ans: 3.5 to 4.5 | ● ○ ○

Weights, initial and final bits are shown in the figure

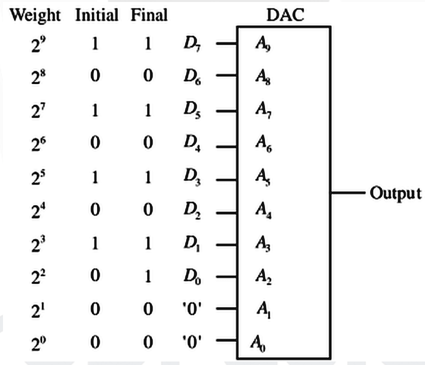


Fig. Solution Q6.20.1

Here, the input data is changed by 1 bit but line change in analog input will be of resolution or 1 step size.
Change in DAC output = 4x the step size (as weightage of bit changed is 2^2

$$\text{Step size} = \text{Resolution} = \frac{\text{Full scale value}}{2^n - 1}$$

Given 10 bit adc, $n = 10$, and full scale value = 1.023 V.

$$\text{Step size} = \text{Resolution} = \frac{1.023}{2^{1}0 - 1} = 1 \text{ mV}$$

$$\therefore\ \text{Change in analog output} = 4 \times \text{step size} = 4 \text{ mV}$$

$$\boxed{4}$$

Q6.20.2 NAT 2020 1M Ans: 223.3 to 223.5

Given:

- Dual Slope ADC
- Full scale reading V_{fs} = 200 mV
- Analog reference voltage V_R = 100 mV
- Applied input analog voltage V_{in} = 123.4 mV

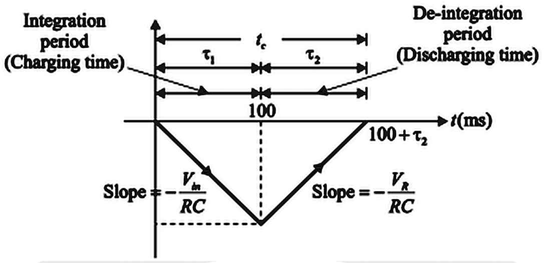


Fig. Solution Q6.20.2

First integration time (charging time) = τ_1 = 100 ms

Input voltage x Charging time = Reference voltage x Discharging time

$$|V_{in} \times \tau_1| = |V_R \times \tau_2|$$

$$123.4 \times 100 \times 10^{-6} = 100 \times 10^{-3}\tau_2$$

$$\tau_2 = 123.4 \text{ ms}$$

where τ_2 = Discharging time (de-integration time)

Total conversion time = Integration time + De-integration time

$$t_c = \tau_1 + \tau_2 = 100 + 123.4 = 223.4 \text{ ms}$$

$$\boxed{223.4}$$

Q6.19.1 NAT 2019 1M Ans: 64

In an N-bit binary weighted resistor DAC,

$$\text{LSB resistance} = 2^{N-1}\times\text{MSB resistance}$$

$$\text{MSB resistance} = 2^{N-1} \times 0.5 \text{ k}\Omega = 2^7 \times 0.5 \text{ k}\Omega = 64 \text{ k}\Omega$$

$$\boxed{64}$$

Q6.18.1 MCQ 2018 1M Ans: C

Number of bits $n = 8$

n bit flash ADC requires 2^n-1 comparators.

$\therefore$ number of comparators needed is = 2^8-1 = 255

$$\boxed{(C)\ 255}$$

Q6.14.1 MCQ 2014 1M Ans: A

Quantisation noise power = $\dfrac{\Delta^2}{12}$, where Δ = step size. Step size is defined as:

$$\Delta = \frac{V}{2^N - 1}$$

$$\therefore \text{Quantisation noise power} = \frac{\Delta^2}{12} = \frac{V^2}{12(2^N - 1)^2}$$

$$\boxed{(A)\ \frac{V^2}{12(2^N - 1)^2}}$$

Q6.14.2 MCQ 2014 1M Ans: A

Maximum error in ADC is step size , i.e. , resolution.

$$\text{Step size} = \frac{V_{ref}}{2^n - 1}$$

$$\therefore \text{Error} \propto V_{ref}$$

$$\boxed{(A)\ \text{directly proportional to } V_{ref}}$$

Q6.11.1 MCQ 2011 1M Ans: C

Given:

- 4-bit SAR ADC with input range of 0 to 15 volts
- Output bit b_1 next to the LSB has a stuck-at-zero fault

Checking for option A: For 2V and 4V the expected step wise conversion is:

For 2V				For 4V			
b_3	b_2	b_1	b_0	b_3	b_2	b_1	b_0
1	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0

2 V and 4 V show different values.

Checking for option B: For 4V and 6V the expected step wise conversion is:

For 4V				For 6V			
b_3	b_2	b_1	b_0	b_3	b_2	b_1	b_0
1	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0
0	1	0	0	0	1	0	0
0	1	0	0	0	1	0	1

4 V and 6 V show different values.

Checking for option C: For 1V and 2V the expected step wise conversion is:

For 1V				For 2V			
b_3	b_2	b_1	b_0	b_3	b_2	b_1	b_0
1	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1

1 V and 2 V show same values.

Checking for option D: For 8V and 9V the expected step wise conversion is:

For 8V				For 9V			
b_3	b_2	b_1	b_0	b_3	b_2	b_1	b_0
1	0	0	0	1	0	0	0
1	0	0	0	1	1	0	0
1	0	0	0	1	0	0	0
1	0	0	0	1	0	0	1

8 V and 9 V show different values.

$$\boxed{(C)\ 1\text{ V and }2\text{ V}}$$

Q6.14.2 NAT 2014 1M Ans: 0.5

Since opamp is in negative feedback, virtual short is valid, i.e. , voltages at inverting and non-inverting terminals are same. Therefore,

$$V^- = V^+ = 0\text{ V}$$

Applying KCL at inverting terminal,

$$\frac{0 - V_{o2}}{R} + \frac{0 - V_{o1}}{8\text{ k}\Omega} + \frac{0 - V_o}{8\text{ k}\Omega} = 0$$

$$\frac{V_o}{8\text{ k}\Omega} = \frac{-V_{o1}}{8\text{ k}\Omega} - \frac{V_{o2}}{R} \quad ----(i)$$

Now, since V_{o1} and V_{o2} are DAC outputs, they can be written as:

$$V_{o1} = -V_{ref}[b_0 + 2b_1 + 4b_2 + 8b_3] \quad ----(ii)$$

$$V_{o2} = -V_{ref}[b_4 + 2b_5 + 4b_6 + 8b_7] \quad ----(iii)$$

From (i) , (ii) , (iii) , we obtain:

$$\frac{V_o}{8\text{ k}\Omega} = \frac{V_{ref}[b_0 + 2b_1 + 4b_2 + 8b_3]}{8\text{ k}\Omega} + \frac{V_{ref}[b_4 + 2b_5 + 4b_6 + 8b_7]}{R}$$

$$V_o = V_{ref}\left(b_0 + 2b_1 + 4b_2 + 8b_3 + \frac{8\text{ k}\Omega[b_4 + 2b_5 + 4b_6 + 8b_7]}{R}\right)$$

$$\frac{8\text{ k}\Omega}{R} = 2^4$$

$$R = 0.5\text{ k}\Omega$$

$$\boxed{0.5}$$

Q6.10.1 MCQ 2010 1M Ans: A

Given: $n = 4$, $V_{FS} = 15$ V

Step size of ADC is given by

$$\text{Step size} = \frac{V_{FS}}{2^n - 1} = \frac{15}{2^4 - 1} = 1 \text{ V}$$

MSB is set and analog voltage is compared to $(1000)_2 = (8)_{10}$ equivalent to 8 V which is less than 8.15 V, so this bit remains set. The SAR process is as follows:

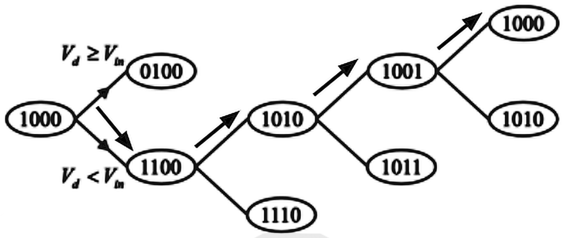


Fig. Solution Q6.10.1

$$\boxed{(A)}$$

Q6.09.1 MCQ 2009 1M Ans: D

Given:

- Nominal input (V_i) range is -2 V to +2 V , i.e. , $V_{max} = 2$ V and $V_{min} = -2$ V
- Number of bits $n = 8$
- It generates a digital code of 00 H for an analog input in the range -7.8125 mV to +7.815 mV, which implies that 0 V is coded as 00 H.

Resolution/step size is given by:

$$\text{Step size} = \frac{V_{max} - V_{min}}{2^n - 1} = \frac{4}{2^8 - 1} = 15.6 \text{ mV}$$

To reach -1.5 V , no. of steps required is:

$$\text{Number of steps required} = \frac{1.5}{15.6 \times 10^{-3}} = 95.62 \approx 96$$

Therefore the digital output is $-(96)_{10}$.

$$\text{2's complement of } -(96)_{10} = 1010\ \ 0000 = \text{A0 H}$$

$$\boxed{(D)\ \text{A0 H}}$$

Q6.08.1 MCQ 2008 1M Ans: C

Given:

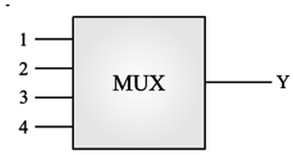


Fig. Solution Q6.08.1

Sequence of output Y is 1 , 2 , 1 , 3 , 1 , 4 , 1 , 2 , 1 , 3 , 1 , 4 , 1 , ..., and each output is generated after 5 μs.
In the sequence, 1 is repeated every 10 μs. Therefore, T_s = 10 μs.
Sampling frequency is given by:

$$f_s = \frac{1}{T_s} = \frac{1}{10\ \mu s} = 100 \text{ k samples/s}$$

$$\boxed{(C)\ 100\text{k samples/s}}$$

Q6.08.2 MCQ 2008 1M Ans: B

Given:

- $n = 12$
- Conversion time = 5 μs

$$\text{Quantization Error} = \frac{\pm\Delta}{2}$$

where Δ = Step size or Resolution.

$$\text{Resolution} = \frac{V_{FS}}{2^n - 1}$$

$$\%\ \text{Resolution} = \frac{1}{2^{12} - 1} \times 100\%$$

$$\%\ \text{Quantization Error} = \frac{\pm\ \%\ \text{Resolution}}{2} = \frac{1}{2(2^{12} - 1)} \times 100\% = \pm 0.012\ \%$$

$$\boxed{(B)\ \pm\ 0.012\ \%}$$

Q6.08.3 MCQ 2008 1M Ans: D

Conversion time = 5 μs

$$\text{Cutoff Frequency} = \frac{1}{\text{Conversion time}} = \frac{1}{5\ \mu s} = 200 \text{ kHz}$$

$$\boxed{(D)\ f_c = 200 \text{ kHz}}$$

Q6.07.1 MCQ 2007 1M Ans: B ● ○ ○

Given circuit is:

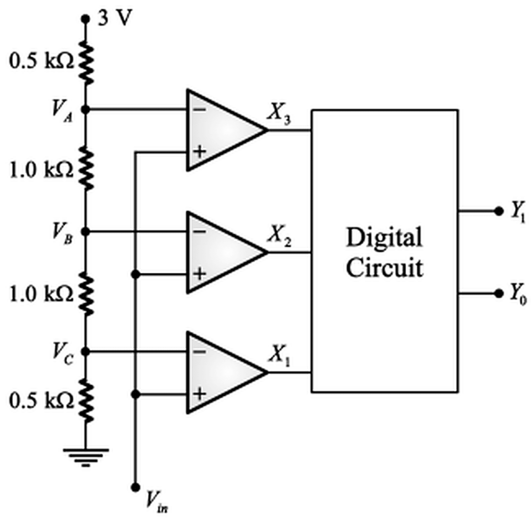


Fig. Solution Q6.07.1

Applying voltage division rule,

$$V_c = \frac{3 \times 0.5 \text{ k}\Omega}{(1 + 1 + 0.5 + 0.5) \text{ k}\Omega} \implies V_c = 0.5 \text{ V}$$

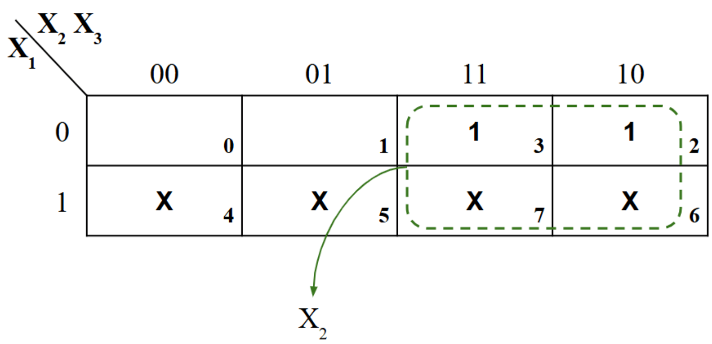
$$V_B = \frac{3 \times (1 + 0.5) \text{ k}\Omega}{(1 + 1 + 0.5 + 0.5) \text{ k}\Omega} \implies V_c = 1.5 \text{ V}$$

$$V_A = \frac{3 \times (0.5 + 1 + 0.5) \text{ k}\Omega}{(1 + 1 + 0.5 + 0.5) \text{ k}\Omega} \implies V_c = 2.5 \text{ V}$$

So the truth table becomes:

V_{in}	X_3	X_2	X_1	Y_1	Y_0
$0 < V_{in} \leq 0.5$ V	0	0	0	0	0
$0.5 < V_{in} \leq 1.5$ V	0	0	1	0	1
$1.5 < V_{in} \leq 2.5$ V	0	1	1	1	0
$2.5 < V_{in}$	1	1	1	1	1

K-Map for Y_1, including unused input combinations as don't-cares:



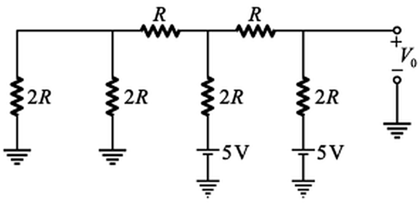
$$\therefore Y_1 = X_2$$

$$\boxed{(B)\ X_2}$$

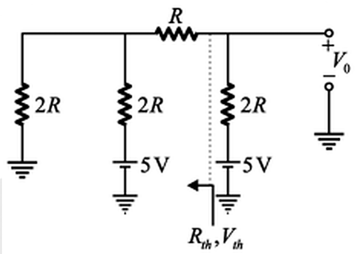
Q6.06.1 MCQ 2006 1M Ans: B ● ○ ○

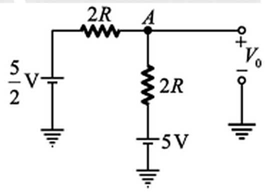
Given: $S_0 = 0$, $S_1 = 1$, $S_2 = 1$ and number of bits $n = 3$

For the given switch combination, the circuit is:



Modified circuit is:





Applying KCL at node A,

$$\frac{V_0 - \frac{5}{2}}{2R} + \frac{V_0 - 5}{2R} = 0 \implies V_0 = \frac{15}{4}$$

$$\boxed{(B)\ \frac{15}{4}\ \text{V}}$$

Q6.06.2 MCQ 2006 1M Ans: D ● ○ ○

Step size of the DAC is given by:

$$\text{Step size } = \frac{V_{FS}}{2^n - 1} = \frac{5}{2^3 - 1} = 0.714 \text{ V} \approx 0.75 \text{ V}$$

$$\boxed{(D)\ 0.75 \text{ V}}$$

Q6.06.3 MCQ 2006 1M Ans: D ● ○ ○

In an ADC along with S-H circuit, to avoid error at output, voltage across capacitor should not drop by more than $\frac{\pm\Delta}{2}$, where Δ is the step size.

$$\Delta = \frac{10 - 0}{2^{10} - 1} = 9.775 \text{ mV} \implies \frac{\pm\Delta}{2} = 4.8875 \text{ mV}$$

So, maximum conversion time should be such that the drop across capacitor voltage must reach maximum value $\frac{\Delta}{2}$. The time can be calculated as:

$$t = \frac{\frac{\Delta}{2}}{\text{Drop Rate}} = \frac{4.8875 \text{ mV}}{10^{-4} \text{ V/ms}} \approx 49 \text{ ms}$$

$$\boxed{(D)\ 49 \text{ ms}}$$

Q6.05.1 | MCQ | 2005 | 1M | Ans: D | ● ○ ○

Conversion time of N-bit SAR-ADC is given by

$$T = NT_c$$

where T_c is the clock period.

$$\therefore T \propto N$$

$$\boxed{(D)\ \text{N}}$$

Q6.04.1 | MCQ | 2004 | 1M | Ans: C | ● ○ ○

Number of bits $n = 8$

n bit flash ADC requires 2^n-1 comparators.

$\therefore$ number of comparators needed is = 2^8-1 = 255

$$\boxed{(C)\ 255}$$

Q6.03.1 | MCQ | 2003 | 1M | Ans: B | ● ○ ○

Given:

- Reference voltage $V_R = 100$ mV
- First integration period $T_1 = 50$ ms
- Input voltage $V_{in} = 120$ mV
- R = 100 kΩ , C = 0.04 μF

For a dual-slope ADC, Input voltage x Charging time = Reference voltage x Discharging time

$$V_{in} \times T_1 = V_R \times (T_2 - T_1)$$

where $T_2 - T_1$= De-integration period

$$120 \times 10^{-3} \times 50 \times 10^{-3} = 100 \times 10^{-3} \times (T_2 - T_1)$$

$$(T_2 - T_1) = 60 \text{ ms}$$

$$\boxed{(B)\ 60 \text{ ms}}$$

Q6.99.1 MCQ 1999 1M Ans: B ● ● ○

The given circuit is an R-$2R$ inverted ladder DAC for the input digital code 1100.

The equivalent resistance seen by the reference source is:

$$R_{eq} = 10\,\text{k}\Omega$$

Therefore, the total reference current is:

$$I_T = \frac{10\,\text{V}}{10\,\text{k}\Omega} = 1\,\text{mA}$$

The total current at the inverting terminal of the lower op-amp (summing $I_T/8$ and $I_T/16$ branches) is:

$$I' = \frac{I_T}{8} + \frac{I_T}{16} = \frac{1}{8} + \frac{1}{16} = \frac{3}{16}\,\text{mA}$$

Using the virtual ground condition at the lower op-amp's inverting node:

$$I' = \frac{0 - V_a}{10\,\text{k}}$$

Solving for V_a:

$$V_a = -10\text{k} \times I' = -10\text{k} \times \frac{3}{16}\,\text{mA} = \frac{-30}{16}\,\text{V}$$

Computing current I_1 from node V_b to V_a:

$$I_1 = \frac{V_b - V_a}{10\text{k}} = \frac{-V_a}{10\text{k}} = \frac{3}{16}\,\text{mA}$$

Applying KCL at node V_b:

$$\frac{I_T}{2} + \frac{I_T}{4} + I_2 = I_1$$

$$\frac{1}{2} + \frac{1}{4} + I_2 = \frac{3}{16}$$

$$I_2 = \frac{3}{16} - \frac{3}{4} = \frac{3 - 12}{16} = \frac{-9}{16}\,\text{mA}$$

Using the virtual ground condition at the upper op-amp:

$$I_2 = \frac{V_0 - 0}{10\text{k}}$$

Solving for V_0:

$$V_0 = 10\text{k} \times I_2 = \frac{-90}{16}$$

$$\boxed{V_0 = -5.625\,\text{V} \quad \text{(nearest option)}}$$

Key Points

- In an R-$2R$ inverted ladder DAC, the current through each $2R$ branch is binary-weighted: $I_T/2, I_T/4, I_T/8, I_T/16$, corresponding to the MSB to LSB.

Mistakes to be avoided

- Verify which bits are "ON" (connected to I_T branch) versus "OFF" (grounded) for the specific input code before summing currents at each node.

Q6.99.2 MCQ 1999 1M Ans: D

Given:

- SAR ADC
- Number of bits $n = 8$
- Clock frequency $f_c = 1$ MHz

For SAR ADC, conversion time is fixed and is given as:

$$T = nT_c = \frac{n}{f_c} = \frac{8}{1\text{ MHz}} = 8\ \mu\text{s}$$

$$\boxed{(D)\ 8\ \mu\text{s}}$$

Q6.99.3 MCQ 1999 1M Ans: C

A dual-slope ADC integrates an unknown input voltage V_{in} for a fixed amount of time, then de-integrates using a known reference voltage for a variable amount of time.

The key advantage of this architecture is that the conversion result is **insensitive to errors in the component values**. Any error introduced by a component value (such as the integrating capacitor or clock frequency drift) during the integrate cycle is automatically **cancelled out** during the de-integrate phase, since the same component is used in both phases.

Thus, the dual-slope ADC can significantly reduce errors due to component tolerances and power supply variations.

$$\boxed{(C)\ \text{can reduce errors due to power supply}}$$

Key Points

- Since the same integrating capacitor and clock are used in both phases, any drift or error in these components affects both phases equally and **cancels out** in the final ratio, making the conversion highly accurate.
- This self-cancelling property makes dual-slope ADCs ideal for high-precision, low-speed applications such as digital multimeters, where noise immunity and accuracy matter more than conversion speed.

Mistakes to be avoided

- Do not confuse the dual-slope ADC's error cancellation with speed — dual-slope ADCs are inherently **slow** due to the two-phase integration process; the advantage is accuracy, not speed.
- Do not assume this insensitivity applies to *all* errors; it specifically cancels errors common to both the integrate and de-integrate phases (e.g., capacitor value, clock period), not offset or noise errors unique to one phase.

Q6.98.1 MCQ 1998 1M Ans: A

Given $n = 12$

$$\%\ \text{Resolution} = \frac{1}{2^n - 1} \times 100\% = \frac{1}{2^8 - 1} \times 100\% = 0.39\ \%$$

$$\boxed{(A)\ 0.39}$$

Q6.98.2 MCQ 1998 1M Ans: C

Given:

- Full scale voltage $V_{fs} = 10$ V
- Resolution required $\Delta = 5$ mV

$$\Delta \times \text{Number of steps} = \text{Full scale voltage}$$
$$5 \times 10^{-3} \times \text{Number of steps} = 10 \text{ V}$$
$$\text{Number of steps} = \frac{10}{5 \times 10^{-3}} = 2 \times 10^{3}$$
$$\text{Number of steps} \leq 2^n - 1$$
$$2000 \leq 2^n - 1 \implies n \geq 10.966 \implies n = 11$$
$$\boxed{(C)\ 11}$$

Key Points

- If resolution is given in volts, it means step size (Δ) has been given.

Q6.98.3 MCQ 1998 1M Ans: B

Given $n = 12$ and $f_c = 4$ MHz

$$\text{Charging time } T = 2^n T_c = \frac{2^n}{f_c} = \frac{2^8}{4 \times 10^6} = 0.064 \text{ ms}$$
$$\boxed{(B)\ 0.064 \text{ ms}}$$

Q6.97.1 MCQ 1997 1M Ans: A

Among different types of ADCs, the **successive approximation type ADC** requires a Sample-and-Hold (S/H) circuit at its front end.

This is necessary because the successive approximation algorithm evaluates the input signal **bit by bit** over multiple clock cycles, comparing the input against a sequence of reference voltages generated by an internal DAC. If the input signal changes during this multi-cycle evaluation, the comparison results would be inconsistent, leading to an incorrect digital output.

Therefore, an S/H circuit is astutely critical for a successive approximation quantizer in order to keep the input waveform **stable** during the successive iterations of bit evaluation.

$$\boxed{(A)\ \text{Successive approximation}}$$

Q6.96.1 MCQ 1996 1M Ans: A

Given

- $R = 10$ kΩ
- $n = 3$
- $V_{ref} = 8$ V
- $R_F = 2R$

$$V_{out} = \frac{V_{ref} R_F}{2^n R}[b_0 + 2b_1 + 4b_2] = \frac{8 \times 20}{8 \times 10}[1 + 2 + 4] = 14 \text{ V}$$
$$I_{out} = \frac{V_{out}}{2R} = \frac{14}{20} = 0.7 \text{ mA}$$
$$\boxed{(A)\ 0.7 \text{ mA}}$$

Q6.96.2 MCQ 1996 1M Ans: D ● ○ ○

Given:

- $V_{FS} = 1.275$ V
- $n = 8$

$$\text{Step size } \Delta = \frac{V_{max} - V_{min}}{2^n - 1} = \frac{1.275}{2^8 - 1}$$

$$\text{Quantization error} = \pm\frac{\Delta}{2} = \pm\frac{1.275}{2(2^8 - 1)} = \pm 2.5 \text{ mV}$$

$$\boxed{(D) \ \pm\ 2.5 \text{ mV}}$$

Q6.94.1 NAT 1994 1M Ans: 0.1 ● ○ ○

Given $n = 10$ bits

$$\% \text{ Resolution} = \frac{1}{2^n - 1} \times 100\% = \frac{1}{2^{10} - 1} \times 100\% = 0.0977\ \% \approx 0.1\ \%$$

$$\boxed{0.1}$$

DEBUG INFO

Chapter VI

ADC & DAC

Total Questions : 39

MCQ : 28

MSQ : 1

NAT : 10

SOLUTIONS

VII

Logic Families and Semiconductor Memories

Topics

1. Logic Gate Implementation Using CMOS
2. Characteristics of Logic Families
3. Hazards in Combinational Circuits
4. Programmable Logic Devices (PLD, PLA, PAL)
5. ROM, SRAM and DRAM

Q7.26.1 MSQ 2026 1M Ans: B , C

In the given CMOS circuit, **Pull Down network** has one nmos controlled by A , in parallel with a series combination of nmos transistors controlled by B and C . Thus, pull-down condition is $A + B.C$.

Output is complement of pull-down logic, so

$$f = \overline{A + B.C}$$

which matches (B).

Also, by De-Morgan's Law,

$$f = \bar{A}.(\bar{B} + \bar{C})$$

which matches (C).

$$\boxed{(B)\ f = \overline{A + B.C}}$$

$$\boxed{(C)\ f = \bar{A}.(\bar{B} + \bar{C})}$$

Mistakes to be avoided

- Always verify that both Pull-up and Pull-down networks give the same expression.
- Do not forget to complement the expression obtained from network to get the final output.

Q7.23.1 MCQ 2023 1M Ans: B

Emitter-Coupled Logic is the fastest among the given options.

$$\boxed{(B)\ \text{Emitter-Coupled Logic}}$$

Q7.15.1 MCQ 2015 1M Ans: A

Given:

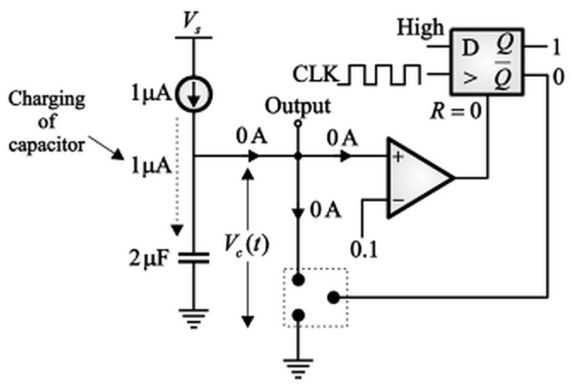
$$R_{ON}(\text{NMOS}) = 1000\,\Omega, \quad R_{OFF}(\text{NMOS}) = \infty, \quad f_{clock} = 1\,\text{Hz}$$

The capacitor voltage is given by:

$$V_c(t) = \frac{1}{C}\int I\,dt$$

Case 1: $V_c(t) < 0.1$ **V**

Circuit is:



Here $R = 0$ (i.e., $Q = 1, \bar{Q} = 0$), so the NMOS is off (open circuit). The current source charges the capacitor linearly:

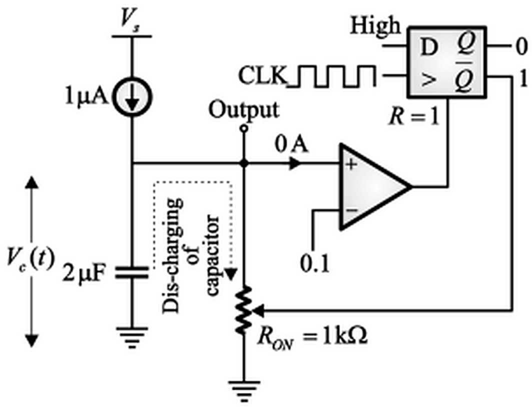
$$V_c(t) = \frac{1}{C}\int_0^t I\,dt = \frac{1}{2\,\mu\text{F}}\int_0^t (1\,\mu\text{A})\,dt$$

$$V_c(t) = \frac{1}{2}t \text{ V}$$

The capacitor charges linearly until $V_c(t)$ reaches 0.1 V.

Case 2: $V_c(t) > 0.1$ **V**

Circuit is:



Here $R = 1$ (i.e., $Q = 0$, $\bar{Q} = 1$), turning the NMOS on. The capacitor discharges rapidly through R_{ON} with time constant:

$$\tau = R_{ON} \times C = 1000 \times 2 \times 10^{-6} = 2 \text{ msec}$$

Since $\tau = 2$ msec is very small compared to the charging time, the discharge is nearly instantaneous, causing the capacitor voltage to drop sharply back to near zero before the slow linear charging cycle repeats.

This produces a characteristic sawtooth-like waveform with sharp discharge spikes.

$\boxed{(A)}$

Key Points

- This circuit implements a relaxation oscillator: a constant current source linearly charges a capacitor, while a comparator with hysteresis (via the flip-flop) triggers rapid discharge once a threshold is reached.
- The charging phase is linear ($V_c \propto t$) because the current is constant; the discharging phase is exponential but extremely fast due to the small RC time constant relative to the charging period.
- The asymmetry between slow linear charging and fast exponential discharging produces the characteristic sawtooth (not triangular or sinusoidal) waveform shape.

Mistakes to be avoided

- Do not assume the discharge is instantaneous; it follows an exponential decay with time constant $\tau = R_{ON}C$, but this τ is much smaller than the charging time, making it *appear* instantaneous on the same timescale.

Q7.14.1 MCQ 2014 1M Ans: D

The truth table for the given logic circuit is given by

A	B	P_1	P_2	N_1	N_2	V_0
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

$$V_0 = \bar{A} + \bar{B} = \overline{A.B}$$

Therefore this circuit implements a NAND gate.

$$\boxed{(D)\ \text{NAND}}$$

Q7.07.1 MCQ 2007 1M Ans: C

The truth table for the given logic circuit is given by

A	B	P_1	P_2	N_1	N_2	V_0
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

$$V_0 = \bar{A}.\bar{B} = \overline{A + B}$$

Therefore this circuit implements a NOR gate.

$$\boxed{(C)\ \text{NOR}}$$

Q7.04.1 MCQ 2004 1M Ans: B

Given: Sink current $I_{OL} = 16\,\text{mA}$, $C = 10\,\text{pF}$.

Method 1

The capacitor voltage is given by:

$$V = \frac{1}{C}\int I\,dt = \frac{It}{C}$$

Solving for the discharge time t:

$$t = \frac{CV}{I_{OL}} = \frac{10 \times 10^{-12} \times 12}{16 \times 10^{-3}} = 7.5\,\text{ns}$$

Method 2

For a TTL NAND gate, if the output transistor is OFF, the output equals logic 1 and the gate behaves as a **source**. If the transistor is ON, the output is logic 0 and the gate behaves as a **sink**.

Given that the sink current of the NAND gate is 16 mA, the output is at logic 0. Using Ohm's law across the pull-up path with supply voltage 12 V:

$$16\,\text{mA} = \frac{12 - 0}{R_p}$$

Solving for R_p:

$$R_p = \frac{12}{16\,\text{mA}} = \frac{3}{4}\,\text{k}\Omega = 0.75\,\text{k}\Omega$$

The time constant for the capacitor discharge is:

$$\tau = R_p C = 0.75 \times 10^3 \times 10 \times 10^{-12} = 7.5\,\text{ns}$$

$$\boxed{(B)\ 7.5}$$

Key Points

- In TTL logic, the output stage acts as a **current sink** when driving a logic 0 (transistor ON, pulling current from the load to ground) and as a **current source** when driving a logic 1 (transistor OFF, pull-up resistor sources current to the load).
- The discharge time for a load capacitance can be computed either directly from $t = CV/I_{OL}$ (charge-based) or via the equivalent resistance $\tau = R_pC$ (RC time constant based) — both methods yield identical results when the sink current is constant.
- $R_p = V_{supply}/I_{OL}$ represents the effective pull-down resistance presented by the saturated output transistor when sinking current.

Q7.04.2 MCQ 2004 1M Ans: C

Given circuit is pseudo-NMOS logic, with always-on PMOS. From pull-down network the output Y is:

$$Y = \overline{AB + CD}$$

Given C = 0 and D = 0,so we have

$$Y = \overline{AB}$$

Truth table is:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

It is observed that only when A = 1 , B = 1, we get Y = 0.

$$\boxed{(C)\ A = 1\ ,\ B = 1}$$

Key Points

- In pseudo-NMOS logic, the PMOS is **always-on** (gate tied to ground) and acts as a weak pull-up resistor, while the NMOS pull-down network implements the logic function; the output is high only when the pull-down network is OFF.

Q7.03.1 MCQ 2003 1M Ans: D

From figure, it is observed that V_1 is connected to PMOS and V_2 is connected to NMOS.

Only when both transistors are on, V_0 will be equal to V_i.

For PMOS to be ON, V_1 must be low, i.e. , $V_1 = -V_{EE}$.

For NMOS to be ON, V_2 must be high, i.e. , $V_2 = +V_{DD}$.

$$\boxed{(D)\ V_1 = -V_{EE} \text{ and } V_2 = +V_{DD}}$$

Q7.03.2 MCQ 2003 1M Ans: A ● ○ ○

Circuit according to the question:

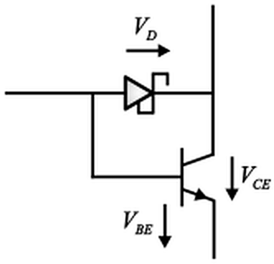


Fig. Solution Q7.03.2

In output transistor of TTL, after introducing schottky diode, it becomes a schottky transistor that prevents the transistor from saturating by diverting the excessive input current. It increases the speed of operation.

$\boxed{(A) \text{ Increases the speed of operation by inhibiting saturation}}$

Q7.94.1 MCQ 1994 1M Ans: A ● ○ ○

Output will be 1 only if both the diodes are reverse-biased. The truth table of the output is:

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

The above table satisfies the AND gate expression.

$\boxed{(A) \text{ AND gate}}$

DEBUG INFO

Chapter VII

Logic Families and Semiconductor Memories

Total Questions : 10

MCQ : 9

MSQ : 1

NAT : 0

SOLUTIONS

VIII

Computer Organization and Architecture

Topics

Q8.26.1 MSQ | 2026 | 2M | Ans: B , C

Given: Each operand is encoded using 3 bits, and the opcode field is 7 bits.

- **Statement (A) is incorrect:** 16 registers would require

$$\log_2 16 = 4\,\text{bits}$$

but only 3 bits are available for each operand.

- The number of unique opcodes representable using a 7-bit opcode field is:

$$2^7 = 128$$

Option (B) is correct.

- The number of internal registers that can be specified using a 3-bit operand field is:

$$2^3 = 8$$

Therefore, the processor has **at most 8** internal registers. **Option (C) is correct.**

- **Statement (D) is incorrect:** 512 opcodes would require

$$\log_2 512 = 9\,\text{bits}$$

while only 7 bits are provided for the opcode field.

$$\boxed{B\ ,\ C}$$

Key Points

- An n-bit field can encode exactly 2^n distinct values; this applies uniformly to register addressing, opcode selection, or any other binary-encoded field.
- The number of bits required to represent M distinct values is $\lceil \log_2 M \rceil$; this is the inverse relationship used to verify whether a given field size is sufficient.

Mistakes to be avoided

- When checking if a claimed value (e.g., 16 registers or 512 opcodes) fits within a given field, always compute $\log_2$(claimed value) and compare it to the available bit-width, rather than checking 2^{bits} against the claimed value loosely.
- Do not assume operand bits and opcode bits are interchangeable; they are separate fields within the instruction word serving different purposes.

Q8.21.1 MSQ | 2021 | 2M | Ans: A , C

16 kB memory = $2^4 k \times 8 = 2^4 \times 10^{10} \times 8 = 2^{14} \times 8 \therefore$ 14 address lines and 8 data lines are required.

A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
1	1	1	1	1	1	1	1	1	1	1	1	1	1	3FFFH

Offset = 3FFFH - 0000H = 3FFFH

- Now if Starting address is 80000H, then

  Ending address = Starting address + Offset = 80000H + 3FFFH = 83FFFH

  Option (A) is correct.

- In Option B, it is given that active high chip select (CS) is needed. Now starting address is F0000H, so ending address will be F0000H + 3FFFH = F3FFFH.

  Chip should be enabled only for addresses F0000H to F3FFFH. If chip select = $A_{19}A_{18}A_{17}A_{16} = 1$, it will enable chip for addresses F0000H to F3FFFH, which is not required.

  Option (B) is incorrect.

- If starting address is 0F000H , then ending address = 0F000H + 3FFFH = 12FFFH.

  For the given address range , we need A_0 to A_{15} address lines. Remaining address lines only can be used for chip select.

  Option (C) is correct.

- Since 8 data lines of processor are available for memory, connection is possible.

  Option (D) is incorrect.

Q8.19.1 | NAT | 2019 | 2M | Ans: 512

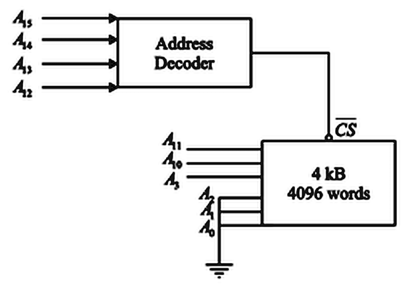


Fig. Solution Q8.19.1

∴ Number of address lines available for main internal access is 12-3 = 9.
∴ Number of addressable words = $2^9 = 512$

$$\boxed{512}$$

Q8.18.1 | NAT | 2018 | 1M | Ans: 279 to 282

Given: 8-bit microprocessor of frequency 5 MHz,

MVIB, 64 H	7T
LOOP : DCR B	4T
JNZ LOOP	10T/7T

The loop is executed 100 times (converting 64H into decimal). For 99 iterations the loop condition is satisfied; on the 100th iteration, the condition is not satisfied, terminating the loop.

The time delay t_{pd} introduced by the program is given by:

$$t_{pd} = 7T + 100(14T) - 3T$$

Simplifying:

$$t_{pd} = 7T + 1400T - 3T = 1404T$$

The clock period T is the reciprocal of the clock frequency:

$$T = \frac{1}{f} = \frac{1}{T_{CLK}} = 0.2\,\mu\text{sec}$$

Substituting into the total delay expression:

$$t_{pd} = 1404 \times 0.2\,\mu\text{sec} = 280.8\,\mu\text{sec}$$

$$\boxed{280.8}$$

Key Points

- Loop delay analysis requires separating instructions executed **once** (outside the loop, contributing a fixed cycle count) from instructions executed **N times** inside the loop body, plus accounting for the final iteration where the exit condition differs.

Q8.17.1 MCQ 2017 1M Ans: B ● ○ ○

Given: 8-bit microcontroller with 16 address lines has 3 fixed interrupts as:

$$\begin{array}{lcl} \text{Int1} & \rightarrow & 0008\text{H} \\ \text{Int2} & \rightarrow & 0010\text{H} \\ \text{Int3} & \rightarrow & 0018\text{H} \end{array}$$

Since the service routine length is more than 8 bytes, it should not disturb vector locations of other interrupts.

$$\boxed{(B)\ \text{An unconditional JUMP to ISS1}}$$

Q8.16.1 MSQ 2016 1M Ans: D ● ○ ○

The given interfacing circuit uses a decoder with select lines S_0, S_1, S_2 connected to address lines A_{12}, A_{13}, A_{14}, and outputs y_0 through y_7, with $\overline{CS}$ activated via output y_5.

The chip should be selected when y_5 is active, which requires:

$$A_{14}A_{13}A_{12} = 101$$

For the decoder's enable inputs: EN_2 to be active requires $A_{11} = 1$, and $\overline{EN_1}$ to be active requires $A_{10} = 0$.

Combining these conditions:

A_{14}	A_{13}	A_{12}	A_{11}	A_{10}
1	0	1	1	0

With $A_{15} = 0$ or $A_{15} = 1$ as the remaining free bit, and all unspecified lower-order address bits (A_9 through A_0) ranging from all 0s to all 1s, the two possible address ranges are:

For $A_{15} = 0$:

$$0\,1\,0\,1\,1\,0\,0\,0\,\cdots 0 = 5800\text{H} \quad \text{to} \quad 0\,1\,0\,1\,1\,0\,1\,1\,\cdots 1 = 5BFF\text{H}$$

For $A_{15} = 1$:

$$1\,1\,0\,1\,1\,0\,0\,0\,\cdots 0 = D800\text{H} \quad \text{to} \quad 1\,1\,0\,1\,1\,0\,1\,1\,\cdots 1 = DBFF\text{H}$$

$$\boxed{(D)\ \text{5800H to 5BFFH and D800H to DBFFH}}$$

Key Points

- The chip-select address decoding logic fixes certain address bits (those connected to decoder select lines and enable inputs) while leaving the remaining bits free; the free bits define the size and range of the addressable memory block.
- Any address bit **not** connected to the decoder (here A_{15} and A_9 through A_0) can take **any value**, producing multiple valid address ranges (aliasing) for the same physical memory chip.
- The size of the address range is determined by the number of free (unconnected) lower-order bits: here 10 free bits (A_9–A_0) give a block size of $2^{10} = 1024$ bytes (400H), matching the range 5800H to $5BFF$H.

Mistakes to be avoided

- Do not assume a single unique address range; unconnected high-order bits (like A_{15} here) create multiple aliased address ranges for the same chip, all of which are valid answers.
- Carefully distinguish between address bits used for **decoding** (fixed values determined by the circuit) and bits left **floating/unused** (free to vary, creating the address range span).
- Double-check the decoder's enable polarity (EN_2 active-high requires $A_{11} = 1$; $\overline{EN_1}$ active-low requires $A_{10} = 0$) — reversing these conditions would shift the address range entirely.

Q8.15.1 MCQ 2015 1M Ans: C

Given: An ADC is interfaced with a microprocessor in the question.

The basic steps in the algorithm for interfacing ADC with microprocessoris shown below:

- Ensuring stability of analog input
- Send SOC (start of conversion) pulse to ADC
- Read the digital data output of ADC

Sending the SOC pulse involves a write cycle ($\overline{WR}$) and to read the digital data output generated by ADC a read cycle is required.

$$\boxed{(C)\text{ One WRITE cycle followed by one READ cycle}}$$

Q8.14.1 MCQ 2014 1M Ans: C

Given: Ext INT1 received. The sequence of operation will be:

- External interrupt device sends request to programmable interrupt controller
- Programmable interrupt controller bypasses that interrupt to microprocessor (INT)
- After recieving interrupt request, microprocessor generates interrupt acknowledgement (INTA)
- Finally programmable interrupt controller sends vector address through data bus to microprocessor

$$\boxed{(C)\text{ Ext INT1} \rightarrow \text{INT} \rightarrow \text{INTA} \rightarrow \text{Address Write}}$$

Q8.11.1 MCQ 2011 1M Ans: C

Given:

- 8K x 8 RAM
- Address of the last memory location = 4FFF H

Method 1

Total memory locations for an 8K × 8 RAM:

$$8\text{K} \times 8\,\text{bits} = 8\text{K}\,\text{bytes} = 8 \times 1024 = 8192\,\text{bytes}$$

Converting 8192 to hexadecimal by repeated division by 16:

$$\begin{array}{r|ll} 16 & 8192 & \\ \hline 16 & 512 & \rightarrow 0 \\ \hline 16 & 32 & \rightarrow 0 \\ \hline 16 & 2 & \rightarrow 0 \\ \hline & 0 & \rightarrow 2 \end{array} \implies \text{Total memory locations} = 2000\text{H}$$

Given final address = 4FFFH. The initial address is found using:

$$\text{Total addresses} = (\text{Final address} - \text{Initial address}) + 1$$

$$\implies \text{Initial address} = \text{Final address} - \text{Total addresses} + 1$$

$$4FFFH - 2000H = 2FFFH$$

$$\text{Initial address} = 2FFFH + 1H = 3000H$$

Method 2

The size of memory for an 8K × 8 RAM is 2000H locations. Given the last memory location is 4FFFH:

$$\text{First memory location} = \text{Last memory location} - \text{Size of memory} + 1$$

$$4FFFH - 1FFFH = 3000H$$

$\boxed{(C)\quad 3000H}$

Key Points

- Memory size in bytes for an $NK \times M$ memory chip is $N \times 1024 \times M/8$ bytes (when M = data width in bits); for byte-addressable RAM, size in addressable locations = $N \times 1024$.
- The relationship between first address, last address, and total size is: Last − First + 1 = Size; rearranging gives First = Last − Size + 1.

Mistakes to be avoided

- Subtracting hexadecimal numbers directly (e.g., 4FFFH−2000H) requires careful hex arithmetic; converting to decimal first can help verify the result if unsure.

Q8.11.2 MCQ 2011 1M Ans: C ● ○ ○

Given: Number of objects crossing a window with variable speed.

Since RST 7.5 is an edge-triggered interrupt, it is best choice of interrupt for counting of objects. The object will be sensed the moment it comes in front of the window. On the other hand, RST 5.5, RST 6.5 and INTR are not suitable since they are level-triggered.

$\boxed{(C)\ \text{RST } 7.5}$

Q8.10.1 MCQ 2010 1M Ans: A ● ○ ○

From figure, output b_2 of 2:4 decoder is used for chip selection of the DAC. Given that $A_{14}A_{13} = 01$, which satisfies the selection of DAC, all other values are taken as zero for initial address and one for final address. The range of memory is given by:

A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	...	A_3	A_2	A_1	A_0	
0	0	1	0	0	.....	0	0	0	0	2000H
0	0	1	1	1	.....	1	1	1	1	3FFFH

From the given options, 3000H is a valid address since it lies within the acceptable address range 2000H to 3FFFH.

$\boxed{(A)\ \ 3000H}$

Q8.10.2 MCQ 2010 1M Ans: C

Execution of the given program is as follows:

Instruction	Operation
MVIA, 00H	Content of A = 00 H
CALL SUB2	Address will be transferred to SUB2
SUB2 : INR A	Content of A = 00H + 1 H = 01H
RET	Address will be transferred to 2005
INR A	Content of A = 01H + 1 H = 02 H
RET	Return to main program.

$\boxed{(C)\ 02}$

Q8.10.3 MCQ 2010 1M Ans: D

Execution of the given program is as follows:

Instruction	Operation
MVI A, 99H	Content of A = 99 H
ADI 11H	Content of A = 99H + 11H = AAH , CY = 0
MOV C, A	Content of C = AAH
RET	Return to main program.

Therefore final content of accumulator is AAH.

$\boxed{(D)\ \text{AAH}}$

Q8.09.1 MCQ 2009 1M Ans: D

DMA stands for direct memory access. It is used to write directly to the RAM without intervention from the processor. During DMA, microprocessor stays idle, i.e., no reading or writing of data occurs.

$\boxed{(D)\ \text{neither reads from nor writes to the bus}}$

Q8.09.2 MCQ 2009 1M Ans: D

Execution of the given program is as follows:

Instruction	Operation
MVI A, 06H	Content of A = 06 H
ADI 70H	Content of A = 06H + 70H = 76H , CY = 0
STA 1007H	Content of location 1007H = 76H

This command replaced code AF (XRA code) with 76H. Therefore XRA is not executed and program moves to HLT, which halts execution. The accumulator contains 76H.

$\boxed{(D)\ \text{76H}}$

Q8.08.1 | MCQ | 2008 | 1M | Ans: B | ● ○ ○

Given: 2K × 8-bit memory interfaced to an 8-bit microprocessor; address of the first memory location = 0800H.

Method 1

Total memory locations for a 2K × 8 memory:

$$2\text{K} \times 8\,\text{bits} = 2\text{K}\,\text{bytes} = 2 \times 1024 = 2048\,\text{bytes}$$

Converting 2048 to hexadecimal:

$$\begin{array}{r|ll} 16 & 2048 & \to 0 \\ \hline 16 & 128 & \to 0 \\ \hline 16 & 8 & \to 8 \\ \hline & 0 & \end{array} \implies \text{Total memory locations} = 0800\text{H}$$

Given the first location address = 0800H. The final address is found using:

$$\text{Total locations} = (\text{Final address} - \text{Initial address}) + 1$$

$$\implies \text{Final address} = (\text{Total locations} + \text{Initial address}) - 1$$

$$= (0800\text{H} + 0800\text{H}) - 1 = 1000\text{H} - 1 = 0\text{FFFH}$$

Method 2

For a 2K × 8 (= $2^{11} \times 8$) memory, 11 address lines (A_{10} to A_0) are required, giving a size of 07FFH. The last memory location is given by:

$$\text{First memory location} + \text{Size of memory}(2\text{K} \times 8)$$

$$0800\text{H} + 07\text{FFFH} = 0\text{FFFH}$$

$$\boxed{(B)\ 0\text{FFFH}}$$

Key Points

- A memory of size Nk × 8 requires $\log_2(N \times 1024)$ address lines; the size in hex address span is $(N \times 1024 - 1)$, i.e., one less than the total location count due to zero-based addressing.
- Standard sizes (in hex span): 1K = 03FFH, 2K = 07FFH, 4K = 0FFFH, 8K = 1FFFH — memorizing these accelerates address range problems.

Q8.07.1 | MCQ | 2007 | 1M | Ans: D | ● ○ ○

From the given table,

- Content of lower and higher address lines are 20H and 20H.
- Data 24H indicates that content of accumulator is 24H.
- $\text{IO}/\overline{\text{M}} \to$ Logic High : Indicates that I/O operation will execute
- $\overline{\text{WR}} \to$ Logic Low : Indicates that write operation will execute

Thus assembly language instruction being executed as OUT 20H.

$$\boxed{(D)\ \text{OUT 20 H}}$$

Q8.07.2 MCQ 2007 1M Ans: A ● ○ ○

The execution of the program proceeds as follows:

Instruction	Operation
IN PORT	$A \leftarrow$ PORT
RAL	**Case 1:** Let the number be positive, i.e., 00000110. After RAL, the MSB of the accumulator rotates into carry: Carry = 0. Since JNC is true, the same process repeats (loop continues).
	Case 2: If the number is negative, i.e., 10000001. After RAL, the MSB rotates into carry: Carry = 1. Since JNC is no longer true, the microprocessor exits the loop.

Thus, the program continuously reads input from the port and rotates the accumulator left, checking the carry (which reflects the MSB/sign bit) after each rotation. The loop continues as long as the input is non-negative (MSB = 0), and **halts** as soon as a **negative number** (MSB = 1) is read.

$\boxed{(A)\quad \text{The program halts upon reading a negative number}}$

Key Points

- The RAL (Rotate Accumulator Left) instruction shifts all bits left by one position, with the MSB moving into the Carry flag and the previous Carry moving into the LSB.

Mistakes to be avoided

- Do not confuse RAL (which rotates the MSB into Carry and Carry into LSB — a 9-bit rotation including Carry) with RLC (which rotates the MSB into both Carry and LSB directly, an 8-bit rotation); the carry flag behavior differs subtly between the two.
- Do not assume the loop continues indefinitely; the Carry flag directly mirrors the sign bit only on the *first* RAL after IN PORT, so the test must occur immediately after rotation.
- Remember that JNC jumps (continues the loop) when Carry = 0, and falls through (exits the loop) when Carry = 1 — the loop exit condition is the *negative* number case, not the positive one.

Q8.06.1 MCQ 2006 1M Ans: A ● ○ ○

In memory mapped input/output technique, memory access instructions are used for IO access also.

LXI H, 00FoH → load the address of memory in HL pair

MOV M,A → load the memory by content of A

Hence the correct option is A.

(A)	LXI H, 00FoH MOV M, A

Q8.06.2 MCQ 2006 1M Ans: C

The execution of the program proceeds as follows:

Instruction	Operation
MVI B, 0AH	Content of B = 0AH
LOOP: MVI C, 05H	Content of C = 05H
DCR C	Decreases the content of C by 1. Hence, content of C = 04H
DCR B	Content of B = 09H
JNZ LOOP	Since $B \neq 0$, the JNZ condition is satisfied, so it jumps to LOOP.

This process continues until the content of B becomes zero. The loop runs for 10 times, and on the last iteration the JNZ condition is **false**.

The total number of T-states required is given by:

$$T_{total} = M + (\text{Count}_{10} \times N) - 3 \quad \cdots (i)$$

where M = T-states outside the loop, N = T-states inside the loop.

From the program execution:

$$M = 7T$$

$$\text{Count} = 0\text{AH} = (10)_{10}$$

$$N = 7T + 4T + 4T + 10T = 25T$$

Substituting into equation (i):

$$T_{total} = 7T + (10 \times 25T) - 3T$$

$$\boxed{(C)\ 254T}$$

Key Points

- The $-3T$ correction accounts for the fact that on the final (10th) iteration, the JNZ condition is **not satisfied**, so the jump instruction takes **fewer T-states** (typically 7T instead of 10T for a successful conditional jump) — the difference of $3T$ is subtracted exactly once.
- Total T-states = M + (loop count $\times$ N) − (T-state savings on final exit); this formula structure is common across all loop-delay calculation problems.
- Each instruction's T-state count must be obtained from the standard 8085 instruction set timing table; MVI = 7T, DCR = 4T, JNZ (taken) = 10T, JNZ (not taken) = 7T.

Mistakes to be avoided

- Do not multiply N by the loop count without subtracting the correction term; failing to account for the reduced T-states on the final, unsuccessful jump leads to an overestimate of T_{total}.
- Do not confuse the JNZ taken (10T) vs. not-taken (7T) timing; the difference ($3T$) is exactly the correction subtracted in equation (i).
- Carefully identify which instructions execute **once** (outside the loop, contributing to M) versus those executing **repeatedly** (inside the loop, contributing to N) before setting up the formula.

Q8.05.1 MCQ 2005 1M Ans: D ● ○ ○

Assume $r_1 = 28\text{H}$ and $r_2 = 68\text{H}$. Performing the sequence of three XOR operations:

Step 1: XOR(r_2, r_1):

$$r_1 = 0010\,1000, \quad r_2 = 0110\,1000$$
$$r_2 \leftarrow r_2 \oplus r_1 = 0100\,0000$$
$$r_2 = 40\text{H}, \quad r_1 = 28\text{H}$$

Step 2: XOR(r_1, r_2):

$$r_1 = 0010\,1000, \quad r_2 = 0100\,0000$$
$$r_1 \leftarrow r_1 \oplus r_2 = 0110\,1000$$
$$r_1 = 68\text{H}, \quad r_2 = 40\text{H}$$

Step 3: XOR(r_2, r_1):

$$r_2 = 0100\,0000, \quad r_1 = 0110\,1000$$
$$r_2 \leftarrow r_2 \oplus r_1 = 0010\,1000$$
$$r_2 = 28\text{H}, \quad r_1 = 68\text{H}$$

After the three XOR operations, $r_1 = 68\text{H}$ (originally r_2) and $r_2 = 28\text{H}$ (originally r_1) — the contents of r_1 and r_2 have been **swapped** without using a temporary register.

$$\boxed{(D)\ \text{Contents of registers } r_1 \text{ and } r_2 \text{ are swapped}}$$

Mistakes to be avoided

- Track which register is being *updated* at each step (the first argument is typically the destination); confusing source and destination leads to incorrect intermediate values.

Q8.05.2 MCQ 2005 1M Ans: D ● ○ ○

Time period of a square wave in the audio frequency range is measured by 8085 microprocessor by feeding the square wave to one of the four interrupts RST 7.5, RST 6.5, RST 5.5, or INTR.

Since only RST 7.5 is a positive edge triggered inteerupt and other given interrupts are level triggered, so for accuracy in measurement of time period, RST 7.5 should be selected.

$$\boxed{(D)\ \text{RST } 7.5}$$

Q8.05.3 MCQ 2005 1M Ans: C ● ○ ○

Given the first location address = 8000H and last location address 9FFFH. The size of memory is found using:

$$\text{Total locations} = (\text{Last memory location} - \text{First memory location}) + 1 = 9\text{FFFH} - 8000\text{H} + 1 = 2000\text{H}$$

Converting to decimal,

$$(2000)_{16} = (8192)_{10}$$

Therefore the RAM can store a total of 8192 bytes.

$$\boxed{(C)\ 8192}$$

Q8.05.4 MCQ 2005 1M Ans: B

In DCR M first the opcode is fetched then the content of memory is fetched (i.e. , memory read) , then content is decremented by 1 and again stored in the same memory location (i.e. , memory write).

Opcode fetch cycle	=	4T
Memory read cycle	=	3T
Memory write cycle	=	3T
Total T state	=	10T

Hence B is the correct option.

$$\boxed{(B)\text{ Op-code fetch, memory read, memory write}}$$

Q8.05.5 MCQ 2005 1M Ans: A

Tracing the execution of the 8085 program step by step:

LXI H, 0000H:

$$HL = 0000\text{H}$$

PUSH H: The content of HL is pushed onto the stack.

$$(200F\text{H}) = 00\text{H}, \quad (200E\text{H}) = 00\text{H}$$

$$SP \leftarrow 200EH$$

PUSH H: HL is pushed again.

$$(200D\text{H}) = 00\text{H}, \quad (200C\text{H}) = 00\text{H}$$

$$SP \leftarrow 200CH$$

POP B: The top of the stack is popped into the BC register pair.

$$B \leftarrow 00\text{H}, \quad C \leftarrow 00\text{H}$$

$$SP \leftarrow 200EH$$

DAD SP: The content of HL is added to SP.

$$HL = 0000\text{H} + 200E\text{H} = 200E\text{H}$$

XCHG: The contents of the HL and DE register pairs are exchanged.

$$HL \leftrightarrow DE$$

$$DE = 200E\text{H}, \quad HL = 1234\text{H}$$

Thus, after execution, $HL = 1234,$ H and $DE = 200E,$ H.

$$\boxed{(A)\text{ 200E H}}$$

Key Points

- **PUSH** decrements SP by 2 and stores the register pair at the new stack location, higher byte first.
- **POP** retrieves the register pair from the stack and increments SP by 2.
- **DAD SP** adds the value of the stack pointer to the HL pair.
- **XCHG** swaps the contents of HL and DE register pairs.

Q8.04.1 MCQ 2004 1M Ans: C

Given: Software interrupt command = RST7

The vector address corresponding to RST7 = $(7 \times 8)_{10} = (56)_{10} = (38)_{16} = 0038$ H

$$\boxed{(C)\ 0038\ \text{H}}$$

Q8.04.2 MCQ 2004 1M Ans: D

The execution of the program proceeds as follows:

Instruction	Operation
XRA A	Content of A = 0
MOV L,A	Content of L = 0
MOV H,L	Content of H = 0
INX H	Increment of the content of H by 1 $\implies$ 0000H + 1 = 0001H
DAD H	Contents of the HL pair = 0001H + 0001H = 0002H

$$\boxed{(D)\ 0002\text{H}}$$

Q8.03.1 MCQ 2003 1M Ans: C

This is a matching-type question pairing 8085 microprocessor signals/terms with their descriptions.

P. RST 7.5: It is a hardware interrupt. It is vectored and maskable, and its vectored address is 003C, H.

$$P \to 3$$

Q. HOLD: The HOLD signal is used for direct memory access.

$$Q \to 4$$

R. IO/M: This signal is used to select I/O or memory operation depending upon its logic level. When $IO/\overline{M} = 0$, the microprocessor is performing a memory-related operation, and when $IO/\overline{M} = 1$, it is performing an I/O-related operation.

$$R \to 1$$

S. ALE: Address Latch Enable (ALE) is a control signal of the 8085 microprocessor, used to demultiplex the lower address and data bus.

$$S \to 2$$

Combining all the correct pairs:

$$P \to 3, \quad Q \to 4, \quad R \to 1, \quad S \to 2$$

$$\boxed{\text{(C) } 3 \quad 4 \quad 1 \quad 2}$$

Key Points

- **HOLD** is used by external devices to request the system bus for direct memory access (DMA).
- $IO/\overline{M}$ distinguishes between memory (0) and I/O (1) operations.

- **ALE** latches the lower-order address from the multiplexed address/data bus.

Q8.03.2 MCQ 2003 1M Ans: D

The execution of the program proceeds as follows:

Instruction	Operation
MVI A, 00H	Content of A = 0
STA 8080H	Content of memory location 8080H = 00H
DCR A	Content of A = 00 - 1 = FFH
STA C080H	Content of memory location C080H = FFH
RET	Return to main program

Therefore Content of memory location 8080H = 00H.

$\boxed{(D)\ 00\text{H}}$

Q8.03.3 MCQ 2003 1M Ans: B

ROM will be active when $\overline{CS}$ is active ,i.e. , when $A_{15} = A_{14} = A_{13} = 0$.

A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	.....	A_3	A_2	A_1	A_0	
0	0	0	0	0	0	0	0	.....	0	0	0	0	0000H
0	0	0	1	1	1	1	1	.....	1	1	1	1	1FFFH

Therefore the address range occupied by ROM is 0000H to 1FFFH.

$\boxed{(B)\ 0000\ \text{-}\ 1\text{FFF}}$

Q8.02.1 MCQ 2002 1M Ans: B

The lower 8 bits (A_0 to A_7) of address register carry both data and address bits, they are time multiplexed, whereas the upper 8 bits carry only address bits.

$\boxed{(B)\ \text{multiplexed}}$

Q8.02.2 MCQ 2002 1M Ans: C

Given:

- 14-bit timer of 8156 loaded with the counter value of $07D0,\text{H}$.
- Input clock frequency = $800,\text{kHz}$.

The 8156 timer is a 14-bit presettable down counter that decrements by 1 for every input pulse.

$$07D0,\text{H} = (2000)_{10}$$

For the first 1000 clock pulses, the timer output remains high, and for the remaining 1000 clock pulses, the output remains low.

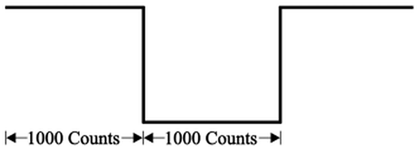


Fig. Solution Q8.02.2

This produces a square wave with a total period corresponding to 2000 clock pulses.
The frequency of the square wave is given by:

$$f_{square} = \frac{1}{2000 \times \frac{1}{800 \times 10^3}} = 400, \text{Hz}$$

$\boxed{(C)\ 400 \text{ Hz}}$

Q8.02.3 MCQ 2002 1M Ans: A

The execution of the program proceeds as follows:

Instruction		Operation
TEST	MOV A, 00H	Content of A = 0
	CALL TESK	Top of stack ← PC (Address of next instruction labelled as TESK) PC ← Address of TESK
TESK	INR A	Content of A = 00H + 1 = 01H
	RET	Top of stack, i.e., Address of the TESK
	INR A	Content of A = 01H + 1 = 02H
	RET	Return to main program

Therefore Content of accumulator = 02H.

$\boxed{(A)\ 02\text{H}}$

Q8.01.1 MCQ 2001 1M Ans: B

m-bit microprocessor can fetch m-bit of opcode from ROM simultaneously and it can execute m-bit operations. For storing m-bit opcodes, the length of instruction register is m-bit. Hence, B is the corrrect option.

$\boxed{(B)\ \text{instruction register}}$

Q8.01.2 MCQ 2001 1M Ans: B

For power on reset of microprocessor there is no supply. In option B it is observed that capacitor behaves as an open circuit for DC signak (0 Hz) .

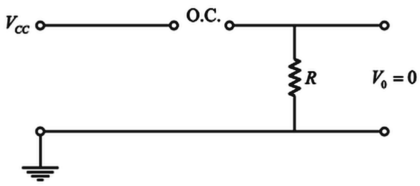


Fig. Solution Q8.01.2

$\boxed{(B)}$

Q8.01.3 MCQ 2001 1M Ans: C

For DMA operation:

1. Once the HOLD signal is received, the microprocessor completes the current machine cycle and generates the HLDA signal. After one clock cycle, it releases the address and data buses to the external device.

2. Once the DMA operation is completed, the HOLD signal becomes low, and the microprocessor then makes the HLDA signal low. After this, one half clock cycle later, the microprocessor takes back (regains) the address and data buses from the external device.

$\boxed{(C) \text{ after half-clock cycle after HLDA goes low}}$

Key Points

- **HOLD** is an input signal requesting the microprocessor to relinquish the buses for DMA operation.
- **HLDA** (Hold Acknowledge) is the output signal confirming that the buses have been released.
- The microprocessor releases the buses one clock cycle after asserting HLDA, and regains them one half clock cycle after HLDA goes low.

Q8.01.4 MCQ 2001 1M Ans: C

If n = number of data lines = 12,

Number of opcodes = $2^n = 2^{12}$

$\boxed{(C)\ 2^{12}}$

Q8.01.5 MCQ 2001 1M Ans: A

This question concerns the effect of various 8085 instructions on the processor flags.

INX and DCX: During the execution of INX and DCX, none of the flags are modified.

CMA: During the execution of CMA (Complement Accumulator), none of the flags are modified.

ANA: The contents of the accumulator are logically ANDed with the contents of a register or memory. During the execution of ANA, the Carry flag is reset and the Auxiliary Carry (AC) flag is set.

SUB: The contents of the operand (register or memory) are subtracted from the contents of the accumulator, and the result is stored in the accumulator. All flags are modified to reflect the result of the subtraction; the Carry flag is the complement of the actual borrow/carry generated.

$\boxed{(A) \text{ SUB}}$

Key Points

- **INX, DCX, CMA** do not affect any flags.
- **ANA** resets the Carry flag and sets the AC flag, while modifying S, Z, P flags based on the result.
- **SUB** affects all flags (S, Z, AC, P, CY), with the Carry flag representing the complement of the borrow.

Q8.00.1 MCQ 2000 1M Ans: D

Given: $SP = F000, \mathrm{H}$, $PC = 2400, \mathrm{H}$.

If the `CALL E000H` instruction is executed, the current contents of the program counter are pushed onto the top of the stack, and the program counter is initialized with $E000, \mathrm{H}$.

Due to the pushing operation, the contents of the stack pointer are decremented by 2.

$$PC = E000, \mathrm{H}$$
$$SP = 23FE, \mathrm{H}$$

$$\boxed{(D)\ \text{PC : E000 , SP : 23 FE}}$$

Q8.00.2 MCQ 2000 1M Ans: A

This question evaluates statements regarding RAM and ROM memory devices.

Option (A): Static RAM uses a completely different technology compared to dynamic RAM. In static RAM, a form of flip-flop holds each bit of memory. A flip-flop memory cell takes 4 to 6 transistors along with some wiring, but it never has to be refreshed. This makes static RAM significantly faster than dynamic RAM. Hence, option (A) is correct.

Option (B): RAM can be used to realize Read Write Memory (RWM). Hence, option (B) is incorrect (the statement itself is true, but it is not the intended answer).

Option (C): ROM is stated to be a type of Random Access Device. Hence, option (C) is incorrect.

Option (D): RAM comes under the category of volatile memory. Hence, option (D) is incorrect.

Hence, the correct option is (A).

$$\boxed{(A)\ \text{Static RAMs are faster than dynamic RAMs}}$$

Q8.00.3 MCQ 2000 1M Ans: B

This is a matching-type question pairing 8085 instructions with their classification/description.

(a) DAA: The contents of the accumulator are changed from a binary value to two 4-bit Binary Coded Decimal (BCD) digits. It is an arithmetic instruction.

a $\rightarrow$(h)

(b) LXI (Load Register Pair Immediate): This instruction loads 16-bit data into the register pair designated in the operand. It is a data movement instruction.

b $\rightarrow$ (f)

(c) RST: It is basically a quick subroutine call to one of 8 predefined addresses (0X00, 0X08, 0X10, 0X18, 0X20, 0X28, 0X38). It is a type of software interrupt instruction.

c $\rightarrow$ (g)

(d) JUMP: It is a branch control or program control instruction.

d $\rightarrow$ (e)

$$\boxed{(B)\ \text{a-h, b-f, c-g, d-e}}$$

Q8.99.1 MCQ 1999 1M Ans: C ● ○ ○

Microcontroller is a small computer on a single integrated circuit. One of its most important features is that it has on-chip interface for I/O devices.

$$\boxed{(C) \text{ interface for I/O devices}}$$

Q8.99.2 MCQ 1999 1M Ans: A ● ○ ○

Given:

- Memory space = 2^{16}
- Word length = 24

For 10 address lines and 8 data lines, number of chips will be

$$N = \frac{2^{16} \times 24}{2^{10} \times 8} = 192$$

$$\boxed{(A)\ 192}$$

Q8.98.1 MCQ 1998 1M Ans: B ● ○ ○

RST 5 comes under the category of software interrupt.

$$\boxed{(B) \text{ RST 5}}$$

Q8.97.1 MCQ 1997 1M Ans: C ● ○ ○

The stack pointer in a microprocessor is a register containing the address of the top of the stack and it works on last in first out principle.

$$\boxed{(C) \text{ the address of the top of the stack}}$$

Q8.94.1 MCQ 1994 1M Ans: A ● ○ ○

Given: 8-bit microprocessor with 16 address lines.

The 8085 microprocessor uses 2's complement arithmetic, and the range of signed integers in 2's complement representation is given by:

$$\text{Range} = (-2^{n-1}) \text{ to } (2^{n-1} - 1)$$

where n is the number of bits.

For an 8-bit number:

$$\text{Range} = -(2^{8-1}) \text{ to } (2^{8-1} - 1) = -128 \text{ to } 127$$

This range is nearest to option (A).

$$\boxed{(A) \text{ -127 to 127}}$$

Q8.94.2 NAT 1994 1M Ans: 16 ● ○ ○

Registers A,B,C,D,E,H,L,W,Z are of 8 bit and PC and SP are of 16 bit. Hence the program counter on an 8-bit microprocessor is always a 16-bit register.

$$\boxed{16}$$

Q8.94.3 MCQ 1994 1M Ans: C ● ○ ○

The storage and retrieval of the content of registers on the stacks follow the LIFO (last in first out) sequence.

$$\boxed{(C)\text{ last in first out basis}}$$

DEBUG INFO

Chapter VIII

Computer Organization and Architecture

Total Questions : 48

MCQ : 42

MSQ : 3

NAT : 3